

GPR30 in spinal cholecystokinin-positive neurons modulates neuropathic pain

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Qing Chen, Hui Wu, Shulan Xie, Fangfang Zhu, Fang Xu, Qi Xu, Lihong Sun, Yue Yang, Linghua Xie, Jiaqian Xie, Hua Li, Ange Dai, Wenxin Zhang, Luyang Wang, Cuicui Jiao, Honghai Zhang, Xuelong Zhou, Zhen-Zhong Xu ✉, Xinzhong Chen ✉

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China • NANHU Brain-Computer Interface Institute, Hangzhou, China • Department of Anesthesiology, The Second Affiliated Hospital, Chongqing Medical University, Chongqing, China • Department of Anesthesiology, Affiliated Hangzhou First People's Hospital, Westlake University School of Medicine, Hangzhou, China • Department of Anesthesiology, Sir Run Run Shaw Hospital, School of Medicine, Zhejiang University, Hangzhou, China • Department of Anesthesiology, First Affiliated Hospital and School of Brain Science and Brain Medicine, Zhejiang University School of Medicine, Hangzhou, China • Liangzhu Laboratory, MOE Frontier Science Center for Brain Science and Brain-machine Integration, State Key Laboratory of Brain-machine Intelligence, NHC and CAMS Key Laboratory of Medical Neurobiology, Zhejiang University, Hangzhou, China

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eLife Assessment

This **important** study investigates nerve-injury-induced allodynia by studying the role of a subpopulation of excitatory dorsal horn CCK⁺ neurons that express the estrogen receptor GPR30 and potentially modulate nociceptive sensitivity via direct inputs from primary somatosensory cortex. In this revised version, the authors addressed many of the critiques raised through added analyses that **convincingly** support the notion that spinal GPR30 neurons are indeed an excitatory subpopulation of CCK⁺ neurons that contribute to neuropathic pain. While evidence of a direct functional corticospinal projection to CCK⁺/GPR30-neurons is not fully demonstrated, this work will be of broad interest to researchers interested in the neural circuitry of pain.

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Abstract

Neuropathic pain, a major health problem affecting 7-10% of the global population, lacks effective treatment due to its elusive mechanisms. Cholecystokinin-positive (CCK⁺) neurons in the spinal dorsal horn (SDH) are critical for neuropathic pain, yet the underlying molecular mechanisms remain unclear. Here, we show that the membrane estrogen receptor G-protein coupled estrogen receptor (GPER/GPR30) in spinal neurons was significantly upregulated in chronic constriction injury (CCI) mice and that inhibition of GPR30 in CCK⁺ neurons reversed CCI-induced neuropathic pain. Furthermore, GPR30 in spinal CCK⁺ neurons was essential for the enhancement of AMPA-mediated excitatory synaptic transmission in CCI mice. Moreover,

GPR30 was expressed in spinal CCK⁺ neurons that received direct projection from the primary sensory cortex (S1-SDH). Chemogenetic inhibition of S1-SDH post-synaptic neurons alleviated CCI-induced neuropathic pain. Conversely, chemogenetic activation of these neurons mimicked neuropathic pain symptoms, which were attenuated by spinal inhibition of GPR30. Finally, we confirmed that GPR30 in S1-SDH post-synaptic neurons was required for CCI-induced neuropathic pain. Taken together, our findings suggest that GPR30 in spinal CCK⁺ neurons is pivotal for neuropathic pain and GPR30 mediates descending facilitation by corticospinal direct projections, thereby representing a promising therapeutic target for neuropathic pain.

Introduction

Neuropathic pain, a persistent and disabling condition affecting approximately 7–10% of the global population, arises from a lesion or dysfunction of the somatosensory nervous system [1–4]. This disorder manifests as mechanical allodynia (pain triggered by innocuous stimuli) and thermal hyperalgesia (exaggerated pain responses to noxious heat) [1, 5, 6]. However, the complex and elusive mechanisms of neuropathic pain still make it difficult to treat. Therefore, identifying precise mechanisms and discovering new therapeutic targets is urgently needed.

The spinal cord (SC) serves as a pivotal hub for processing peripheral sensory inputs and integrating descending modulatory signals. Neuropathic mechanical allodynia occurs via circuit-based transformation in this region [4, 5, 7, 8]. Superficial laminae of the SC primarily encode noxious stimuli, whereas deeper laminae process innocuous signals. Under neuropathic conditions, however, low-threshold mechanosensory inputs aberrantly activate nociceptive neurons in superficial laminae through disinhibition and sensitization mechanisms, driving mechanical allodynia. Excitatory interneurons, constituting ~75% of spinal dorsal horn (SDH) neurons, are central to this pathological process [9–11]. Among these, cholecystokinin-expressing (CCK⁺) interneurons, enriched in SDH deep laminae, have recently been proposed as mediators of both mechanical and thermal hypersensitivity [10, 12]. Notably, CCK⁺ neurons receive direct corticospinal projections from the primary sensory cortex (S1), modulating neuropathic pain sensitivity [13]; however, the molecular underpinnings of this regulation remain obscure.

Estrogen, beyond its classical nuclear receptors (ER α /ER β), exerts non-genomic effects via the membrane receptor G protein-coupled estrogen receptor (GPR30), which is increasingly recognized as a key player in nociceptive modulation [14–26]. Although GPR30's role in pain modulation has been documented, its specific contribution to SDH circuitry in neuropathic pain remains unexplored.

In this study, we demonstrate that spinal GPR30 orchestrates neuropathic pain by modulating excitability of CCK⁺ neurons and corticospinal descending facilitation. We discovered that GPR30 activation in spinal CCK⁺ neurons was both required and sufficient for the development of neuropathic pain. Interestingly, we observed that GPR30 in spinal CCK⁺ neurons was required for the enhancement of spontaneous excitatory post-synaptic currents (sEPSC) in CCI mice, in an AMPA-dependent manner. Importantly, we revealed that GPR30 was expressed in the spinal CCK⁺ neurons receiving direct projections from S1 sensory cortex, and that GPR30 in S1-SDH post-synaptic neurons was critical for CCI-induced neuropathic pain. These findings establish GPR30 in spinal CCK⁺ neurons as a compelling therapeutic target for neuropathic pain management.

Results

Spinal inhibition of GPR30 reverses CCI-induced neuropathic pain and neuronal activation

To investigate the functional relevance of spinal GPR30 in neuropathic pain, we employed a chronic constriction injury (CCI) model to evaluate whether pharmacological blockade of spinal GPR30 alleviates neuropathic pain (**Fig 1A**). Quantitative PCR (qPCR) analysis revealed a significant elevation of *Gper1* (GPR30) mRNA levels in the lumbar spinal dorsal horn (SDH) of CCI mice compared to sham (**Fig 1B**). By the way, the *Gper1* mRNA levels in the DRG remained unchanged after CCI (**Fig 1B**). Intrathecal application of the GPR30 antagonist, G-15, effectively attenuated CCI-induced mechanical allodynia and thermal hyperalgesia in both sexes of mice (**Fig 1C-E**), whereas basal nociceptive thresholds remained unaltered in naïve mice (**Fig S1A** and **B**). Of note, the analgesic effects of G-15 lasted for up to 6 hours after intrathecal application in CCI mice (**Fig S1C**). Immunohistochemical data demonstrated that innocuous tactile stimulation triggered pronounced c-Fos expression, a marker of neuronal activation, predominantly in GPR30⁺ cells within the SDH of CCI mice (**Fig 1F-J**). Critically, G-15 treatment substantially suppressed this c-Fos expression (**Fig 1F-J**), indicating that spinal GPR30 inhibition normalizes both hypersensitivity and neuronal activation in neuropathic pain in both sexes.

We further probed whether GPR30 activation alone could mimic neuropathic symptoms in naïve mice. Intrathecal application of the selective GPR30 agonist, G-1, robustly evoked mechanical allodynia and thermal hypersensitivity, with symptom persistence exceeding 48 hours (**Fig S1D-G**). Consistent with behavioral outcomes, G-1 administration markedly increased c-Fos⁺ neuron density in the SDH, with over 80% of activated cells co-expressing GPR30.

Collectively, these findings establish spinal GPR30 as a critical mediator of sex-independent neuropathic pain development in the CCI model.

GPR30 in spinal CCK⁺ neurons is required for CCI-induced neuropathic pain

To delineate the spinal cellular substrates of GPR30-mediated pain modulation, we first mapped its expression pattern. Our results revealed exclusive GPR30 localization to neuronal populations within the SDH instead of astrocytes or microglial cells via immunohistochemistry (**Fig. S2**). Notably, as an estrogen receptor, GPR30 expression showed no sexual dimorphism (**Fig S2D**). To assess the relative expression of GPR30 in excitatory versus inhibitory neurons, we performed *Ish* and immunostaining with *VGAT*, *Vglut2* and GPR30 (**Fig 2A**). GPR30 was predominantly expressed on the excitatory neurons of SDH (**Fig 2B**). Further characterization in wild-type mice using AAV2/9-Camk2-mCherry labeling also confirmed predominant GPR30 enrichment in SDH excitatory interneurons (**Fig S3**).

Given the established role of CCK⁺ neurons in neuropathic pain and their dominance among SDH excitatory interneurons^[10, 21, 27, 28], we hypothesized that GPR30 within this subset drives nociceptive sensitization. To validate this, we characterized CCK expression in the spinal cord via transgenic models by crossing *CCK-Cre* with *Ai14* reporter mice (**Fig 2C**). *Ish* and immunostaining confirmed that our transgenic mice recapitulated the majority of CCK expression in the deep laminae of the SDH as marked by isolectin B4 (IB4), CGRP and NF200 (**Fig 2D** and **E**). To verify the role of CCK⁺ neurons in allodynia under neuropathic pain, we performed

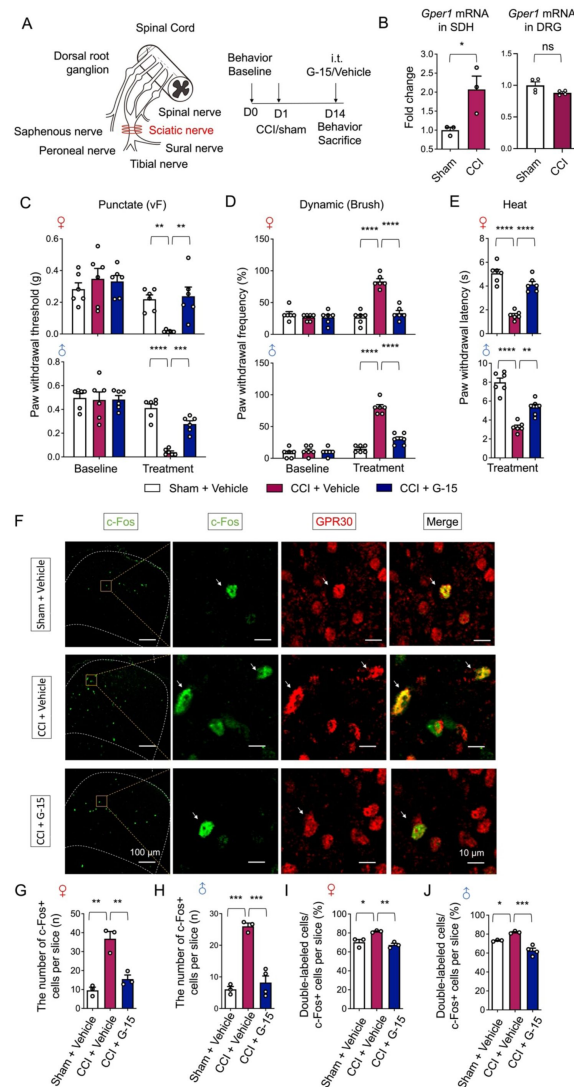


Figure 1.

CCI-induced neuropathic pain and neuronal activation were reversed by spinal inhibition of GPR30.

A Schematic illustration of CCI surgery for induction of neuropathic pain (Left) and diagram showing the timeline of CCI surgery, drug administration and behavioral tests (Right).

B Quantitative PCR analysis of *Gper1* mRNA in SDH from sham and CCI mice (Left, $n = 3$ mice for each group) and *Gper1* mRNA in DRG from sham and CCI mice (Right, $n = 4$ mice for each group).

C-E Behavioral tests of basic nociception and 14 days after CCI or sham surgery along with intrathecal injection of antagonist of GPR30 or vehicle in Von Frey tests (**C**), Brush tests (**D**) and Heat tests (**E**) in mice of both sexes ($n = 6$ mice for each group).

F Immunofluorescence detection of c-Fos (Green) and GPR30 (Red). Scale bars: 100 μm . Boxed area of images is enlarged on the right. Scale bars: 10 μm . White arrows indicate double-positive cells.

G, H Total number of c-Fos positive neurons in the SDH per section in female mice (**G**) and male mice (**H**) ($n = 3-4$ mice for each group, 4-6 pictures were analyzed for each mouse).

I, J Percentage of c-Fos positive neurons expressing GPR30 in female mice (**I**) and male mice (**J**). ($n = 3-4$ mice for each group, 4-6 pictures were analyzed for each mouse).

Data information: in (**B**), $*P < 0.05$ (Unpaired Student's t-test). In (**C, D**), $**P < 0.01$; $***P < 0.001$; $****P < 0.0001$ (two-way ANOVA with Turkey's multiple comparisons test). In (**E**), $**P < 0.01$; $****P < 0.0001$ (one-way ANOVA with Turkey's multiple comparisons test). In (**G-J**), $*P < 0.05$; $**P < 0.01$; $***P < 0.001$ (one-way ANOVA with Turkey's multiple comparisons test). All data are presented as mean $\pm$ SEM.

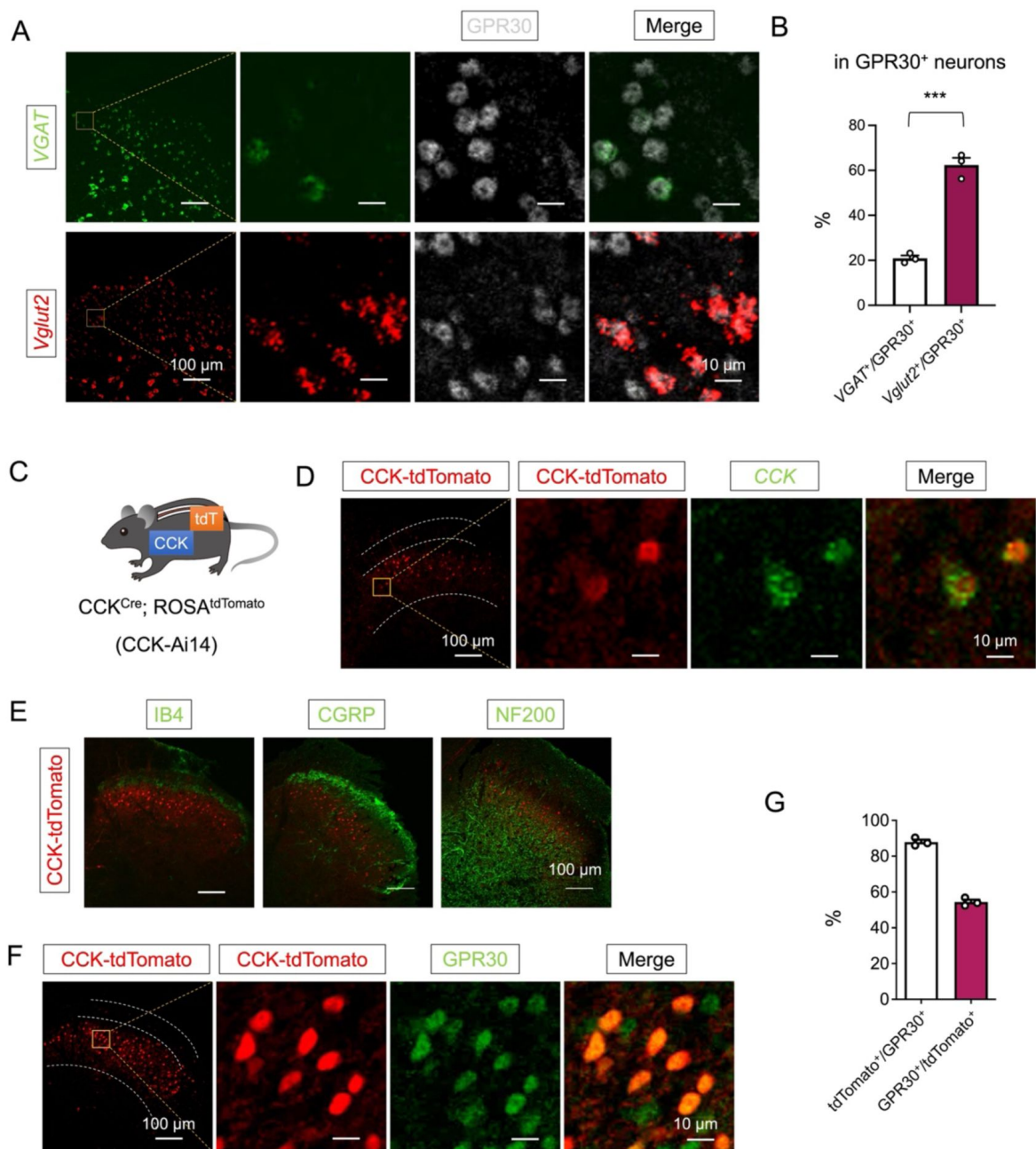


Figure 2.

GPR30 was widely expressed in the CCK⁺ excitatory interneurons in the SDH.

A In situ hybridization showing *VGAT* (Green) and *Vglut2* (Red) with *GPR30* (Gray) in the SDH. Scale bar: 100 μ m. Boxed area of image is enlarged on the right. Scale bars: 10 μ m.

B Percentage of *GPR30* positive neurons expressing *VGAT* or *Vglut2* in wild mice. ($n = 3$ mice, 3-4 pictures were analyzed for each mouse)

C CCK^{Cre} mice express *tdTomato* in a Cre-dependent manner.

D In situ hybridization showing *CCK* (Green) with *tdTomato* (Red) in the SDH. Scale bar: 100 μ m. Boxed area of image is enlarged on the right. Scale bars: 10 μ m.

E Double staining of *tdTomato*⁺ neurons (Red) with *IB4*, *CGRP* or *NF200* (Green) by immunocytochemistry. Scale bars: 100 μ m.

F Immunofluorescence in SDH to verify the co-localization of *tdTomato*⁺ neurons (Red) with *GPR30* (Green). Scale bars: 100 μ m. Boxed area of images is enlarged on the right. Scale bars: 10 μ m.

G Percentage of double labeled neurons in *CCK-tdTomato* and *GPR30* positive neurons. ($n = 3$ mice, 3 pictures were analyzed for each mouse)

immunostaining of c-Fos in CCK-Ai14 mice, and we confirmed that more CCK⁺ neurons were activated after CCI (**Fig S4**). What's more, near-universal CCK⁺ neurons were co-localized with GPR30 (**Fig 2F** and **G**).

To functionally interrogate GPR30 and CCK⁺ neurons in neuropathic pain, we injected AAV2/9-DIO-shGper1-EGFP into lumbar SDH of *CCK-Cre* mice for conditional knock-down of *Gper1*, and 3 weeks later we subjected mice to CCI as well as subsequent behavioral tests (**Fig 3A**). Viral targeting accuracy was verified by immunostaining showing deep laminae-specific expression, mirroring endogenous CCK⁺ neurons distribution (**Fig 3B**). CCI-induced *Gper1* mRNA upregulation was significantly blunted in knockdown mice (**Fig 3C**, **Fig S5A**), confirming the transcriptional efficacy of virus.

Behavioral tests revealed that *Gper1* knockdown in CCK⁺ neurons attenuated CCI-induced mechanical allodynia and thermal hyperalgesia across sexes, without altering basal nociceptive thresholds (**Fig 3D-F**). To dissect affective components of neuropathic pain, we used real-time PEA test with innocuous mechanical stimulation as shown before^[25] (**Fig 3G**). Scramble-treated CCI mice exhibited persistent aversion to stimulated side, whereas *Gper1* knockdown abolished this aversive behavior during post-stimulation period (**Fig 3H-L**).

GPR30 in spinal CCK⁺ neurons is essential for the enhancement of excitatory synaptic transmission in the SDH of CCI mice

To investigate the role of GPR30 expressed on CCK⁺ neurons in neuropathic pain transmission, we conducted electrophysiological recordings using whole-cell patch clamp recording on fluorescently labeled neurons in spinal slices from CCI or sham-operated *CCK-Cre* mice treated with shRNA (**Fig 4A** and **B**). Since a significant proportion of spinal CCK⁺ neurons form excitatory synapses with upstream neurons, we focused on recording postsynaptic currents from GFP-labeled CCK⁺ neurons in laminae III-V approximately two weeks after CCI surgery. As hypothesized, the amplitude of excitatory postsynaptic currents (EPSCs) in CCK⁺ neurons was significantly enhanced in CCI mice compared to sham mice, although the frequency remained unchanged (**Fig 4C-E**). Notably, this enhancement in amplitude was abolished in *Gper1* knockdown mice (**Fig 4C, E**). It is worth mentioning that inhibitory postsynaptic currents (IPSCs) also contribute to the regulation of neuronal excitability. However, our findings indicated that knockdown of *Gper1* did not affect the IPSCs of spinal CCK⁺ neurons in CCI mice (**Fig S5B-D**).

Given that EPSCs are primarily mediated through glutamatergic receptors such as AMPA receptor, coupled with emerging evidence showing that GPR30 could facilitate excitatory transmission via increasing the clustering of glutamatergic receptor subunits^[25], we proceeded to examine whether GPR30-mediated EPSCs depend on AMPA-mediated currents. Electrophysiological experiments were carried out in spinal slices from *CCK-Cre* mice treated with intraspinal injections of AAV2/9-DIO-EGFP (**Fig 4F**). To record AMPA-dependent currents, electrical stimulation from the electrode placed in the deep laminae of SDH was applied (**Fig 4G**). And CCK⁺ neurons were held at -70 mV in the presence of APV (100 μ M) to block NMDA receptors and bicuculline (20 μ M) to block GABA_A receptors and strychnine (0.5 μ M) to block glycine receptors. All recorded cells responded to the electrode stimulation, exhibiting evoked EPSCs (**Fig 4H** and **I**). Furthermore, the application of 0.1 μ M G-1 augmented the amplitude of AMPA currents (**Fig 4H** and **I**). These findings collectively suggest that GPR30 regulates EPSCs in CCK⁺ neurons through an AMPA-dependent mechanism.

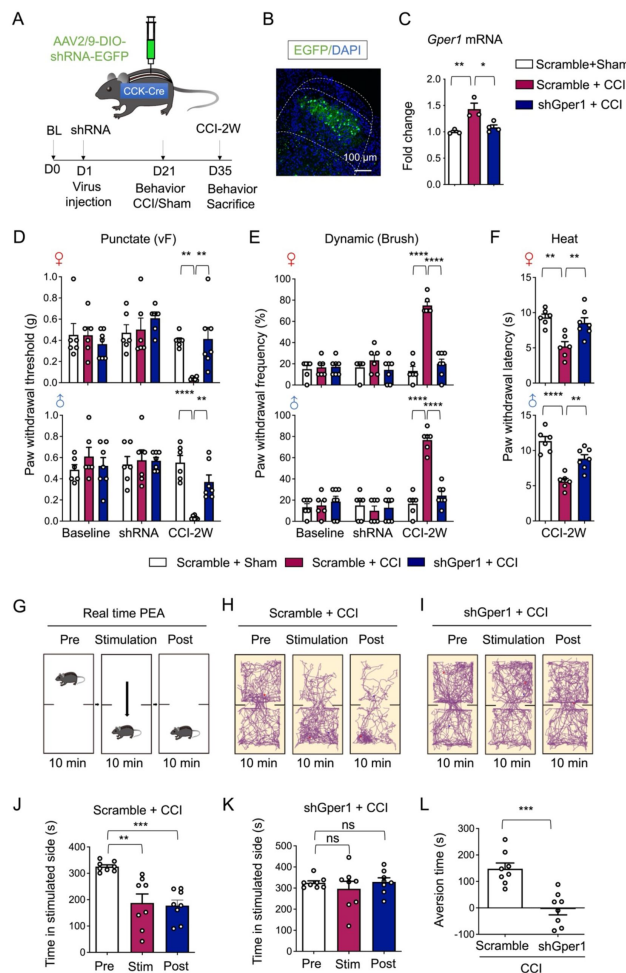


Figure 3.

Knockdown of GPR30 in spinal CCK⁺ neurons alleviated neuropathic pain in CCI mice.

A Schematic illustration of the strategy to knock down Gper1 in the spinal CCK⁺ neurons in a Cre-dependent manner (Up) and the diagram showing the timeline of virus injection, CCI surgery and behavioral tests (Down).

B Immunochemical detection of the localization of virus expression (Green). Scale bars: 100 μ m.

C Quantitative PCR analysis of Gper1 mRNA in SDH from sham and CCI mice with intraspinal virus injection (n = 3-4 mice for each group).

D-F Behavioral tests of basic nociception, 21 days after spinal virus injection and 14 days after CCI or sham surgery along with intrathecal injection of antagonist of GPR30 or vehicle in Von Frey tests (**D**), Brush tests (**E**) and Heat tests (**F**) in mice of both sexes (n = 6-7 mice for each group).

G Schematic illustration of real-time place escape/avoidance (PEA) test.

H, I Representative spatial tracking maps showing the location of a Scramble + CCI group mice (**H**) and a shGper1 + CCI group mice (**I**) before, during and after mechanical stimulation of hind paw in the chambers.

J Quantification of time the Scramble + CCI group mice spent in preferred chamber before, during and after stimulation (n = 8 mice).

K Quantification of time the shGper1 + CCI group mice spent in preferred chamber before, during and after stimulation (n = 8 mice).

L Quantification of the aversion time of mice. (time spent in preferred chamber after stimulation minus the time spent before stimulation) (n = 8 mice for each group).

Data information: in (**C**), *P < 0.05; **P < 0.01 (one-way ANOVA with Turkey's multiple comparisons test). In (**D, E**), ***P < 0.01; ****P < 0.0001 (two-way ANOVA with Turkey's multiple comparisons test). In (**F**), **P < 0.01; ****P < 0.0001 (one-way ANOVA with Turkey's multiple comparisons test). In (**J, K**), *P < 0.05; **P < 0.01; ***P < 0.001; ns = not significant. (one-way ANOVA with Turkey's multiple comparisons test). In (**L**), ***P < 0.001 (Unpaired Student's t-test). All data are presented as mean $\pm$ SEM.

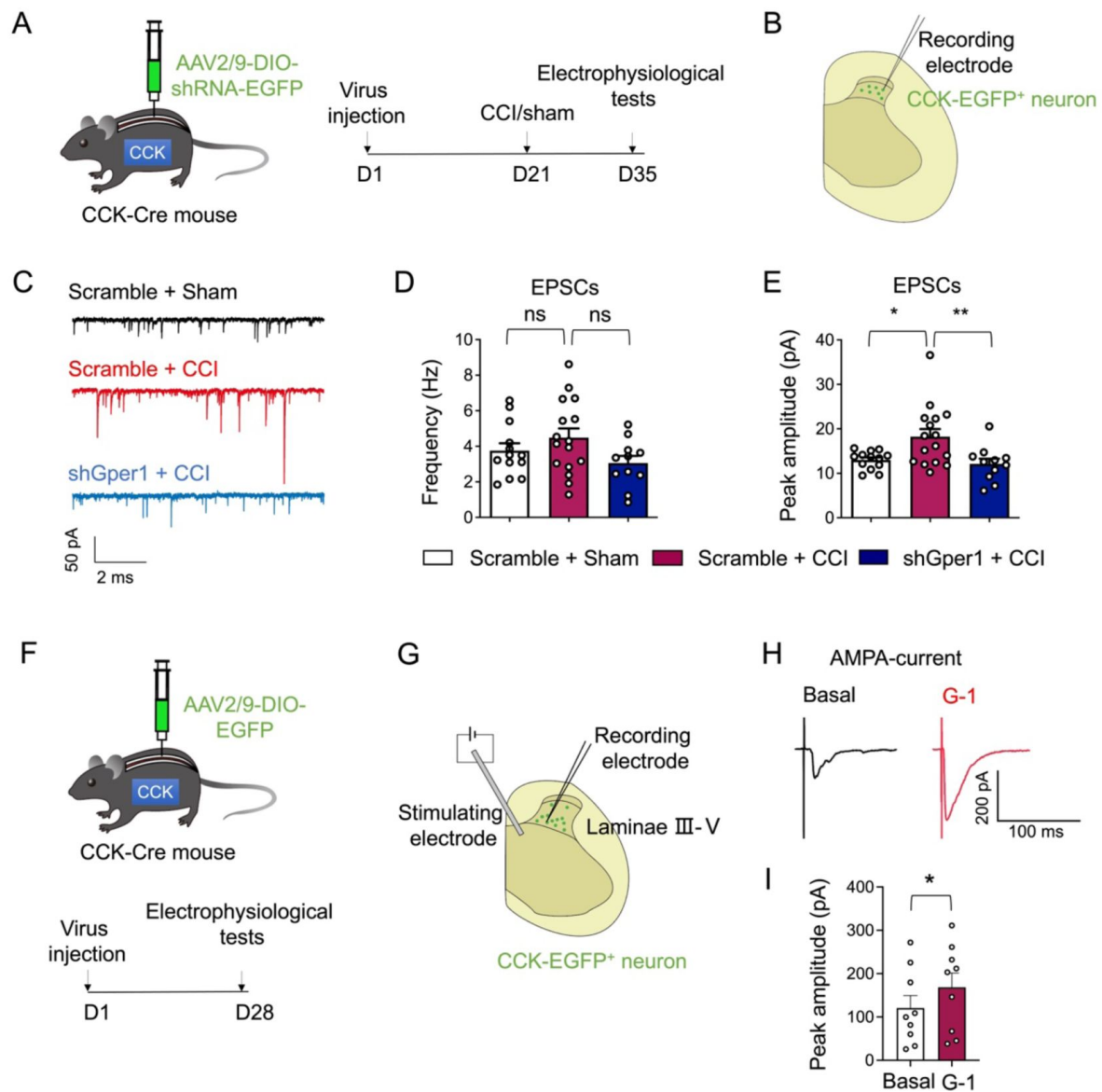


Figure 4.

Knockdown of GPR30 in spinal CCK⁺ neurons reverse the enhancement of sEPSC in CCI mice

A Schematic illustration of the strategy to knock down Gper1 in the spinal CCK⁺ neurons in a Cre-dependent manner (Left) and the diagram showing the timeline of virus injection, CCI surgery and electrophysiological tests (Right).

B Schematic illustration of spinal slice electrophysiological recordings from CCK-EGFP neurons located in the deep laminae of SDH.

C Representative spontaneous EPSCs in CCK-EGFP neurons.

D The frequency of spontaneous EPSCs.

E The peak amplitude spontaneous EPSCs (n = 10-16 cells from 5 mice per group).

F Schematic illustration of the strategy to visualize CCK⁺ neurons in a Cre-dependent manner (Up) and the diagram showing the timeline of virus injection and electrophysiological tests (Down).

G Schematic illustration of spinal slice electrophysiological recordings from CCK-EGFP⁺ neurons located in the deep laminae of SDH during electric stimulation of the deep laminae of SDH to simulate APMA currents.

H Representative AMPA-current in CCK-EGFP⁺ neurons before and after administration of G-1 (0.1 μ M).

I The peak amplitude AMPA currents before and after administration of G-1 (n = 8 cells from 4 mice).

Data information: in (**D**, **E**), *P < 0.05; **P < 0.01; ns = not significant (one-way ANOVA with Turkey's multiple comparisons test). In (**I**), *P < 0.05 (Paired Student's t-test). All data are presented as mean $\pm$ SEM.

GPR30 is expressed in the spinal CCK⁺ neurons receiving direct projection from S1 cortex

To anatomically verify that the lumbar spinal cord receives direct descending projections from the S1 cortex, we employed a combination of retrograde and anterograde tracing methods to investigate the connections between cortical neurons and their axonal innervation in the SDH. Our results revealed that retrograde labeling with cholera toxin subunit B-555 (CTB-555), injected into the deep laminae of the lumbar SDH, predominantly localized neurons in the contralateral S1 cortex (**Fig 5A-C**). Of note, the neurons projecting from S1 cortex are primarily excitatory pyramidal neurons located in layer V^[29]. Similarly, fiber tracing using AAV2/9-hSyn-EGFP indicated that axons from S1 terminate extensively in laminae III-V of the contralateral SDH (**Fig 5D-F**).

Building on previous findings suggesting a functional interaction between S1-SDH projections and spinal CCK⁺ neurons^[13], our current study aimed to further elucidate the structural relationship among GPR30, S1-SDH projections and CCK⁺ neurons. To achieve this, we performed anterograde trans-monosynaptic tracing by injecting AAV2/1-hSyn-Cre into the right S1 cortex, followed by AAV2/9-DIO-mCherry into the contralateral lumbar SDH. This approach allowed us to visualize the mCherry⁺ S1-SDH postsynaptic neurons in the deep laminae of the lumbar SDH (**Fig 5G** and **H**). Consistent with prior observations, we found that these SDH downstream neurons exhibited colocalization with CCK⁺ neurons, with 28.1% of mCherry⁺ neurons expressing CCK (**Fig 5I** and **J**). Thus, our findings confirm that spinal CCK⁺ neurons are directly innervated by the S1 cortex.

To further examine the interplay between GPR30, S1-SDH projections, and CCK⁺ neurons, we utilized a monosynaptic anterograde tracing strategy. Specifically, we injected AAV2/1-EF1α-FLP into the right S1 cortex and AAV2/9-hSyn-Con/Fon-GFP into the contralateral lumbar SDH of CCK-Cre mice. This strategy enabled us to visualize GFP-positive S1-SDH postsynaptic CCK⁺ neurons (**Fig. 5K** and **L**). Our co-staining results revealed that the vast majority of CCK⁺ S1-SDH postsynaptic neurons expressed GPR30 (**Fig 5M** and **N**). These data collectively indicate that the majority of CCK⁺ neurons receiving S1 projections express GPR30.

GPR30 in S1-SDH post-synaptic neurons is critical for CCI-induced neuropathic pain

Given that GPR30 has been verified to be expressed on CCK⁺ neurons receiving S1-SDH direct projections (**Fig 5M**), we employed a combination of chemogenetic and pharmacological approaches to determine whether neurons innervated by S1-SDH direct projections mediate nociception via GPR30^[30]. Specifically, we performed injections of anterograde AAV2/1-hSyn-Cre into the right S1 WT mice. One week later, we administered AAV2/9-hSyn-DIO-hM3Dq (Gq)-mCherry into the lumbar SDH (**Fig 6A**). Pharmacological activation of these post-synaptic neurons with CNO in Gq-treated mice significantly induced spontaneous pain-like behaviors, such as paw scratching, biting, and licking, in the hind paws and tails (**Fig 6B**). Furthermore, chemogenetic activation of S1-SDH post-synaptic neurons dramatically induced mechanical allodynia and thermal hyperalgesia in both sexes compared to negative control groups (**Fig 6C-E**). Notably, the reduction in nociceptive thresholds could be effectively reversed by intrathecal administration of G-15 (**Fig 6C-E**). The corresponding immunohistochemistry results using c-Fos confirmed the chemogenetic activation of S1-SDH post-synaptic neurons, which could be suppressed by intrathecal application of G-15 (**Fig 6F-H**).

To further explore the role of S1-SDH post-synaptic neurons in the modulation of neuropathic pain, we employed chemogenetic inhibitory methods to suppress these neurons in CCI mice (**Fig S6A**). The suppression of S1-SDH post-synaptic neurons could dramatically relieve the mechanical allodynia and thermal hyperalgesia induced by CCI (**Fig S6B-D**). To establish the

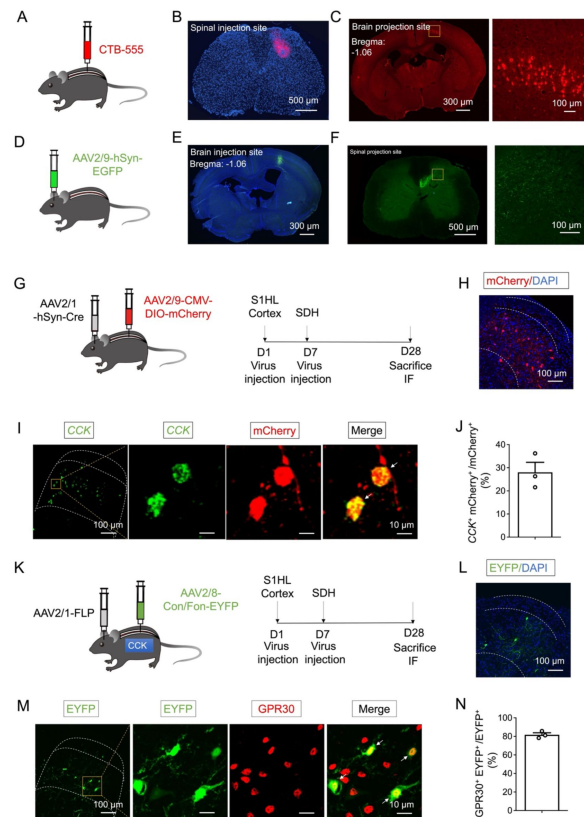


Figure 5.

GPR30 was expressed in the spinal CCK⁺ neurons receiving direct projection from S1 cortex.

A Schematic illustration of the strategy to retrograde tracing the projections innervating spinal dorsal horn via CTB-555.

B Fluorescence image showing the site of CTB-555 injection in the spinal dorsal horn. Scale bars: 500 μ m.

C Coronal brain section showing the location of tdTomato⁺ neurons in the S1HL cortex. Scale bars: 300 μ m. Boxed area of image is enlarged on the right. Scale bars: 100 μ m.

D Schematic illustration of the strategy to anterograde tracing the projections from S1HL cortex.

E Coronal brain section showing the site of AAV injection in the S1HL cortex. Scale bars: 300 μ m.

F Fluorescence image showing the S1HL-SDH tract terminates in the deep laminae of spinal dorsal horn. Scale bars: 500 μ m. Boxed area of image is enlarged on the right. Scale bars: 100 μ m.

G Schematic illustration of the strategy for identifying the spinal post-synaptic neurons of S1-SDH projections (Left) and diagram showing the timeline of AAV2/1 injection in the S1HL cortex, AAV2/9 injection in the spinal dorsal horn and immunofluorescence (Right).

H Representative image showing the mCherry⁺ post-synaptic neurons of the S1HL-SDH projections in the deep laminae of SDH. Scale bars: 100 μ m.

I In situ hybridization showing CCK (Green) with RFP (Red) by immunofluorescence in the spinal dorsal horn. Scale bars: 100 μ m. Boxed area of image is enlarged on the right. Scale bars: 10 μ m. White arrows indicate double-positive cells.

J About 28.13% of RFP⁺ post-synaptic neurons of S1HL-SDH projections expressing CCK (n = 3 mice, 3 pictures were analyzed for each mouse).

K Schematic illustration of the strategy for identifying the spinal CCK⁺ post-synaptic neurons of S1HL-SDH projections in CCK^{Cre} mice (Left) and diagram showing the timeline of AAV2/1 injection in the S1HL cortex, AAV2/9 injection in the spinal dorsal horn and immunofluorescence (Right).

L Representative image showing the EGFP⁺ CCK⁺ post-synaptic neurons of the S1-SDH projections in the spinal dorsal horn with DAPI. Scale bars: 100 μ m.

M Double staining of EGFP⁺ neurons with GPR30 by immunohistochemistry. Scale bars: 100 μ m. Boxed area of image is enlarged on the right. Scale bars: 10 μ m.

N About 81.7% of CCK⁺ post-synaptic neurons of S1HL-SDH projections expressing GPR30 (n = 3 mice, 7-18 pictures were analyzed for each mouse).

Data information: in (**J**, **N**), data are presented as mean $\pm$ SEM.

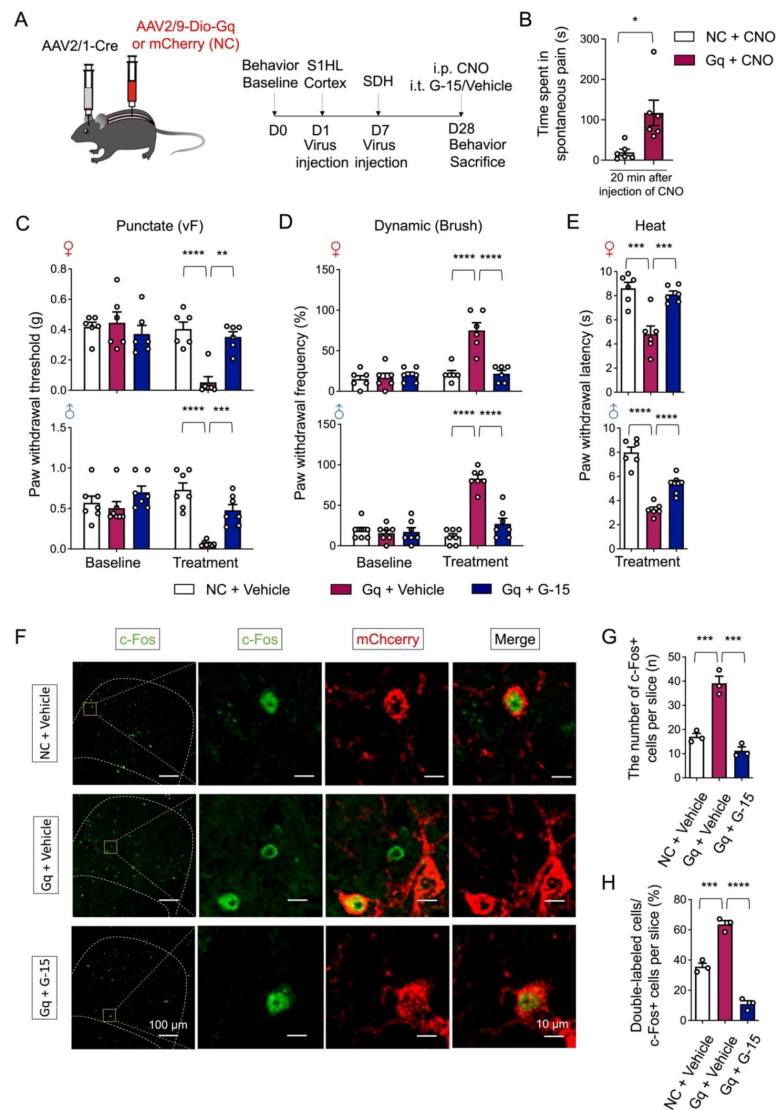


Figure 6.

Chemogenetic activation of S1-SDH post-synaptic neurons mimicked neuropathic pain symptoms, which were reversed by spinal inhibition of GPR30.

A Schematic illustration of the strategy for identifying the post-synaptic neurons of S1HL-SDH projections in SDH (Left) and diagram showing the timeline of AAV2/1 injection in the S1HL cortex, AAV2/9 injection in the spinal dorsal horn and behavioral tests (Right).

B Spontaneous pain induced by intraperitoneal injection of CNO within 20 min ($n = 6$ mice for each group).

C-E Behavioral tests of basic nociception, 28 days after brain virus injection along with intraperitoneal injection of CNO and intrathecal injection of antagonist of GPR30 or vehicle in Von Frey tests (**C**), Brush tests (**D**) and Heat tests (**E**) in mice of both sexes ($n = 6$ mice for each group).

F Immunofluorescence detection of c-Fos (Green) and m-Cherry+ post-synaptic neurons (Red). Scale bars: 100 μm . Boxed area of images is enlarged on the right. Scale bars: 10 μm .

G Total number of c-Fos positive neurons in the SDH per section ($n = 3$ mice for each group, 3 pictures were analyzed for each mouse). **H** Percentage of c-Fos positive neurons expressed in m-Cherry ($n = 3$ mice for each group, 3 pictures were analyzed for each mouse).

Data information: in (**B**), $*P < 0.05$ (Unpaired Student's t-test). In (**C**, **D**), $**P < 0.01$; $***P < 0.001$; $****P < 0.0001$ (two-way ANOVA with Turkey's multiple comparisons test). In (**E**), $***P < 0.001$; $****P < 0.0001$ (one-way ANOVA with Turkey's multiple comparisons test). In (**G**, **H**), $***P < 0.001$; $****P < 0.0001$. (one-way ANOVA with Turkey's multiple comparisons test). All data are presented as mean $\pm$ SEM.

essential role of GPR30 in this process, we specifically knocked down the expression of *Gper1* on S1-SDH post-synaptic neurons and subjected mice to CCI after adequate viral expression (**Fig 7A**). Interestingly, the knockdown of *Gper1* in S1-SDH post-synaptic neurons was sufficient to relieve mechanical allodynia and thermal hyperalgesia in both sexes (**Fig 7B-D**). Immunohistochemistry showed the viral location in deep laminae of SDH (**Fig 7E**) and qPCR confirmed the suppression of *Gper1* mRNA expression which was increased by CCI (**Fig 7F**). Collectively, these findings underscore the critical role of GPR30 in the descending facilitation of S1-SDH direct projections in the neuropathic pain.

Discussion

CCK⁺ neurons, located in the deep laminae of the spinal cord, have long been recognized for their pivotal role in the development and maintenance of neuropathic pain, as well as descending facilitation by sensory cortex-spinal cord projections. Despite extensive research, the molecular mechanisms underlying nociception remain poorly understood. GPR30, as a membrane estrogen receptor, exerts modulatory effects on various physiological and pathological processes, including the development of neuropathic pain, within seconds to minutes^[22]. In this study, we observed significant upregulation of GPR30 expression in the SDH after CCI, underscoring its critical role in neuropathic pain development. To investigate this further, we employed a comprehensive approach, including transgenic mice models, behavioral assays, pharmacological interventions, chemogenetic methods, and electrophysiological studies. These experiments revealed that GPR30 plays an indispensable role in neuropathic pain, particularly through its regulation of AMPA-dependent EPSCs in spinal CCK⁺ neurons, which are essential for pain signaling. Furthermore, we demonstrated that GPR30 is crucial for the descending facilitation of pain signals via corticospinal direct projections. These findings collectively suggest that GPR30 in spinal CCK⁺ neurons may represent a promising therapeutic target for neuropathic pain.

Estrogen has long been recognized as a key player in modulating nociception, with its drastic fluctuations notably impacting nociceptive thresholds^[31]. Our prior research demonstrated that moderate estrogen supplementation effectively mitigated hyperalgesia in ovariectomized (OVX) mice, however, excessive estrogen supplementation paradoxically exacerbated hyperalgesia^[31]. This discrepancy might stem from the fact that varying estrogen concentrations can differentially activate estrogen receptors, including nuclear receptors (ER α and ER β) and membrane receptors (GPR30)^[22]. Considering our findings that inhibiting spinal GPR30 does not alter the basal nociception in naïve mice, it appears that GPR30 may not be significantly activated by estrogen under normal conditions. Additionally, estrogen has long been implicated in modulating neuropathic pain. Intra-dorsal root ganglion (DRG) administration of estrogen in CCI rats has been shown to enhance mechanical and thermal pain in an ER α -dependent manner^[32]. Conversely, estrogen supplementation in the anterior cingulate cortex (ACC) of CCI mice significantly alleviated neuropathic hyperalgesia in a GPR30-dependent manner^[33]. These findings further underscore the distinct roles of different estrogen receptors in pain modulation.

GPR30 is widely expressed in the nervous system and exerts vital effects in nociceptive modulation^[16, 23, 24, 33–37]. For example, the activation of GPR30 in DRG could aggravate the hyperalgesia in OVX mice, while inhibition of GPR30 relieved hyperalgesia^[38]. Besides, the GPR30 expressed on GABAergic cells in rostral ventromedial medulla (RVM) mediates the descending facilitation of nociception^[24]. However, though GPR30 is also widely expressed in SDH^[26], still little is known about the functions as well as underlying mechanisms of spinal GPR30 in nociceptive modulation. Consistent with previous studies^[39], here we found that intrathecal injection of G-1 could dramatically induce mechanical allodynia and thermal hyperalgesia in mice. To further explore whether spinal GPR30 is involved in pathological nociception, we subjected mice to CCI surgery to mimic neuropathic pain^[2, 40]. In accordance with our expectations, the inhibition of spinal GPR30 significantly reversed the mechanical

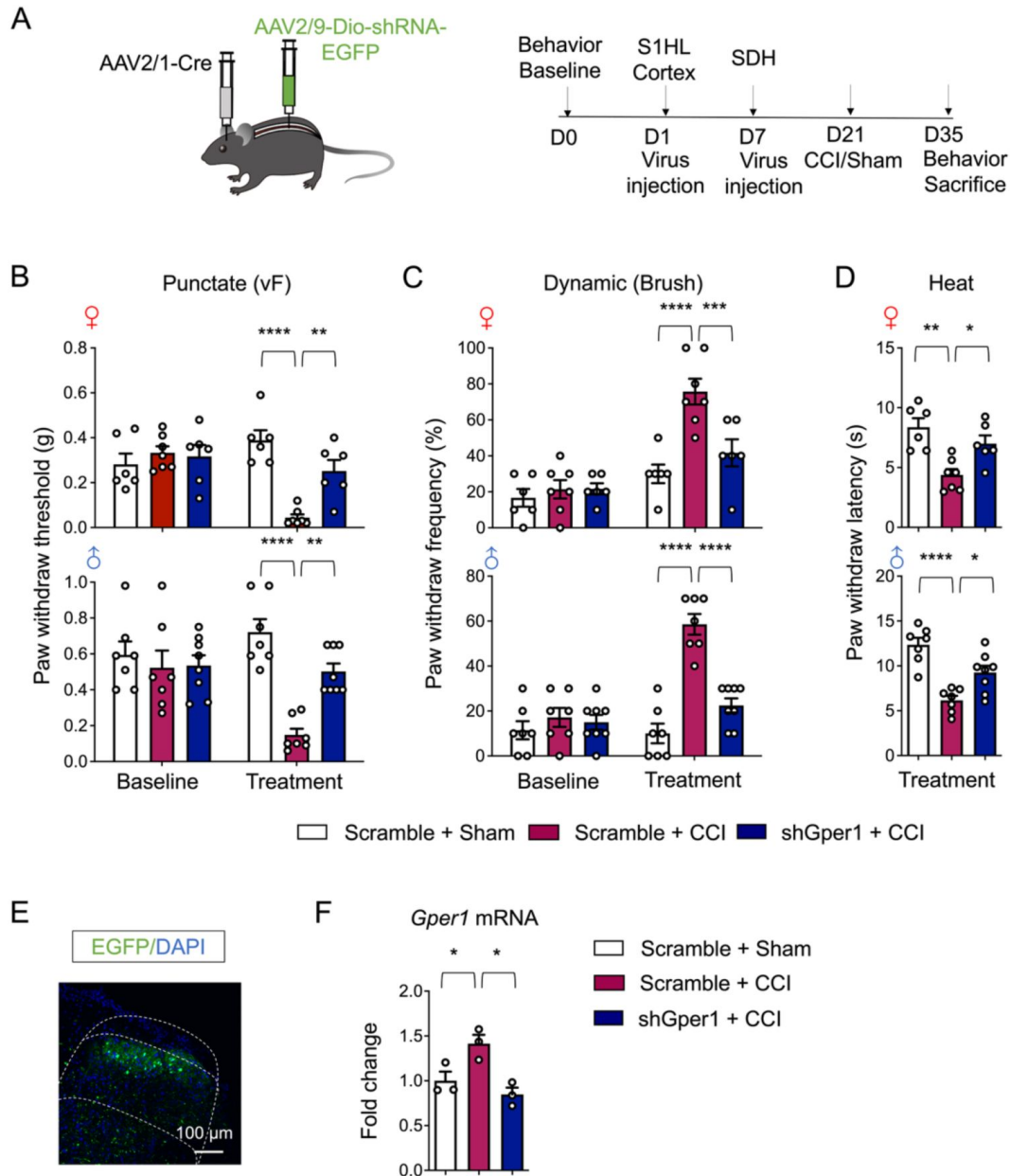


Figure 7.

CCI-induced neuropathic pain was attenuated by knock-down of GPR30 in S1-SDH post-synaptic neurons.

A Schematic illustration of the strategy to knock down Gper1 in the post-synaptic neurons of S1HL-SDH projections in a Cre-dependent manner (Up) and the diagram showing the timeline of virus injection, CCI surgery and behavioral tests (Down).

B-D Behavioral tests of basic nociception and 14 days after CCI surgery in Von Frey tests (**B**), Brush tests (**C**) and Heat tests (**D**) in mice of both sexes (n = 6 mice for each group).

E Immunofluorescence detection of the localization of virus expression (Green). Scale bars: 100 μ m.

F Quantitative PCR analysis of Gper1 mRNA in SDH (n = 3 mice for each group).

Data information: in (**B**, **C**), **P < 0.01; ***P < 0.001; ****P < 0.0001 (two-way ANOVA with Turkey's multiple comparisons test). In (**D**), *P < 0.05; **P < 0.01; ****P < 0.0001 (one-way ANOVA with Turkey's multiple comparisons test). In (**F**), *P < 0.05 (one-way ANOVA with Turkey's multiple comparisons test). All data are presented as mean $\pm$ SEM.

allodynia and thermal hyperalgesia induced by CCI. Moreover, GPR30 has been reported to be involved in nociceptive sexual dimorphism. For example, the regulatory role of GPR30 in DRG in maintenance of hyperalgesia induced by repeated exposure of opioid only exists in female rats^[41, 42]. However, according to our results, spinal GPR30 modulated nociception in a sex-independent manner. We also found that GPR30 is indiscriminately expressed on the neurons of both sexes of mice, which might account for the sex-independent function of spinal GPR30 in nociceptive modulation. Consistent with our results, several studies also have confirmed that the spinal *Gper1* expression showed no significant difference between male and female mice^[43]. What's more, the fluctuation of estrogen failed to change the basic expression of spinal GPR30^[31]. These results further indicate that spinal GPR30 modulated nociception in a sex-independent manner.

The SDH is a major locus for the integration of peripheral sensory input and supraspinal modulation. Most peripheral nociceptive afferents project to the superficial laminae of the SDH which respond to the noxious stimulations, while low-threshold mechanoreceptors form synaptic contacts in the deep laminae which respond to innocuous stimulations^[8]. However, under the condition of mechanical allodynia, innocuous stimulation might also activate more superficial nociceptive circuits and lead to painful perception, which might come from the circuit-based transformation in the SDH^[6, 44]. It should be noted that the spinal dorsal horn is composed of a large number of excitatory (75%) and inhibitory (25%) interneurons, as well as a small part of projection neurons which relay integrated information to various supraspinal regions^[9]. Excitatory interneurons have been confirmed to take a vital role in conveying mechanical allodynia according to the nature of injury^[10, 45]. As a distinct type of excitatory interneurons mainly located in the deeper laminae of SDH, CCK⁺ neurons are important for neuropathic injuries^[10, 12]. The inhibition of spinal CCK⁺ neurons could alleviate neuropathic mechanical allodynia to a great extent^[10]. By the way, spinal CCK⁺ neurons also account for the thermal hyperalgesia^[12]. However, little is known how CCK⁺ neurons mediate the nociception. Combined with our results that GPR30 is widely expressed in spinal excitatory interneurons and involved in neuropathic pain modulation, we speculate that CCK⁺ neurons might convey neuropathic hyperalgesia via GPR30. As expected, most CCK⁺ neurons express GPR30 and knock-down of the *Gper1* in CCK⁺ neurons dramatically relieve the pain induced by CCI, thus indicating the vital role of GPR30 in CCK⁺ neurons mediating neuropathic pain. However, it should be noted that half of GPR30⁺ neurons are not co-localized with CCK⁺ neurons, and further studies are needed to explore the function of these GPR30⁺CCK⁻ neurons in neuropathic pain.

Abnormal activation of neurons in SDH is one of the causes of hyperalgesia and the change of post synaptic currents is the vital factor influencing neuronal excitability^[46, 47]. Given this critical link between EPSCs and excitability, we measured excitatory postsynaptic currents (EPSCs) and found an elevation of EPSCs amplitudes in spinal CCK⁺ neurons after CCI, indicating the increased excitability of CCK⁺ neurons in CCI mice. Furthermore, the knock-down of *Gper1* in CCK⁺ neurons could inhibit the increase of EPSCs amplitude induced by CCI. Together, these data illustrate that GPR30 take a vital role in activation of CCK⁺ neurons after nerve injury via promoting the enhancement of EPSCs. It should be noted that EPSCs are specifically produced by glutamatergic receptors expressed on post-synaptic membrane, including AMPA and NMDA receptors^[25, 46, 48]. In our study, we confirmed that the selective activation of GPR30 by G-1 remarkably enhanced the AMPA-current in spinal CCK⁺ neurons, which might account for the increased excitability of CCK⁺ neurons in neuropathic pain. It should be noted that the IPSCs could also influence the excitability of neurons, however, the knockdown of *Gper1* failed to change the IPSCs amplitude in CCI mice, suggesting that GPR30 did not take part in the inhibitory synaptic regulation.

Increasing evidence has mapped neural circuits from peripheral to central nervous system to illustrate the neural mechanisms of nociception^[5, 7, 9, 45, 49]. In brief, pain is derived from the activation of peripheral nociceptors whose cell bodies lie in DRG, and then nociceptive

signals are transduced to the SDH for preliminary regulation and finally projected to cerebral cortex via a series of brain region mediating nociception. Additionally, pain is also modulated by the descending modulatory pathways constituted of projections from ventrolateral periaqueductal gray (PAG) to the RVM and then to the spinal cord which takes several steps^[50–55]. However, A recent study has come up with the existence of long direct projections from S1 cortex to deep laminae of SDH and the vital role of S1-SDH projections in neuropathic pain^[13]. Inhibition of S1-SDH projections attenuates neuropathic pain, while activation decreases pain thresholds in naive mice. We confirmed the existence of these direct projections and their postsynaptic targets' critical role in neuropathic pain. Notably, these long projections specifically synapse within SDH deep laminae onto CCK⁺ neurons^[13]. We also structurally verify that CCK⁺ neurons receive direct projections from S1 cortex. Furthermore, we also found that the majority of CCK⁺ neurons receiving S1-SDH projections express GPR30, thus indicating an important role of GPR30 in descending modulation of S1-SDH. As expected, the knockdown of *Gper1* in S1-SDH post-synaptic neurons dramatically alleviated the hyperalgesia induced by CCI. All these results further confirmed an important role of GPR30 in descending facilitation of neuropathic pain. However, since viral strategy only labeled a small fraction of post-synaptic CCK⁺ neurons from S1 cortex and was insufficient to functionally manipulate these neurons, more efficient methods should be employed to verify the role of GPR30 expressed on S1-SDH post-synaptic CCK⁺ neurons under neuropathic pain conditions.

Conclusion

GPR30 plays a critical role in the development of neuropathic pain, particularly within CCK⁺ neurons. Our research highlights GPR30's essential function in enhancing AMPA-dependent EPSCs, which are crucial for the activation of CCK⁺ neurons and the subsequent development of abnormal nociception under neuropathic conditions. Furthermore, GPR30 is pivotal in facilitating the descending projections via corticospinal projections, contributing to the propagation of neuropathic pain signals. These findings suggest that targeting GPR30 in spinal CCK⁺ neurons could be a promising therapeutic strategy for neuropathic pain in clinic.

Materials and methods

Animals

Mice of both sexes ranging in age from 8 weeks to 12 weeks were used for this study, including C57BL/6J wild-type (purchased from SLAC Laboratory Animal CO. LTD, Shanghai, China), *CCK-Cre* mice and *Ai14* mice (originally purchased from Jackson Laboratory). In accordance with the Jackson Laboratory's protocol, transgenic mice were genotyped. All animals were kept in a humidity-controlled room with free access to food and water, the facility was maintained at 22 °C and ran on 12 hours of light/dark cycles. A random assignment of animals to different experiment groups was conducted. The animals were treated in accordance with protocols approved by the Animal Ethic and Welfare Committee of Zhejiang University School of Medicine, and all experimental procedures were carried out in accordance with the National Institute of Health Guide for Care and Use of Laboratory Animals (NIH Publications NO.86-23).

Drug administration

For pharmacological manipulation of the activity of spinal GPR30, G-1 or G-15 (diluent of 0.2 mg/mL, administration of 100 µg/kg, 10 µL per mice; APExBIO, USA) was dissolved in 1% DMSO with Saline and administered intrathecally as previously described^[39]. To be specific, mice were lightly anesthetized with 1.5% inhaled isoflurane, and held with a pen under the pelvis while a 25-gauge needle attached to a 10-µL syringe (Hamilton, Nevada, USA) was inserted in the subarachnoid space between vertebrae L5 and L6 until a tail flick was observed. The syringe was held for 30 seconds after the injection of 10 µL solution per mice. For chemogenetic manipulation

of S1-SDH post-synaptic neurons, Clozapine N-oxide (CNO; 2.5 mg/kg, 150 μ L per mice; MCE, China) was dissolved in saline with gentle vortex for mixing and then administered intraperitoneally [30]. The behavioral assessments were carried out 30 minutes following the injection.

Virus and CTB microinjection

For intracranial injection, mice were anesthetized with 1% pentobarbital sodium solution (70 mg/kg per mice) and then secured in a stereotaxic frame (RWD Life Science, Shenzhen, China). A middle scalp incision exposed the skull, and then a hole was drilled on the skull above the right S1 cortex to allow passage of a glass microelectrode filled with the virus. Viral injections were performed with the following coordinates of S1: 0.95~1.15 mm from bregma, 1.4~1.6 mm from midline, and 0.9~1.1 mm ventral to skull. A volume of 300 nL virus was injected at 50 nL/min with calibrated glass microelectrodes by a microsyringe pump (#78-8710 KD Scientific, USA). After infusion, the micropipette was slowly removed after five minutes. For spinal cord injection, mice were anesthetized with 1% pentobarbital sodium solution. The spinal cord could be visible between T12 and T13 vertebral spines following a middle incision along the lumbar vertebrae. With a stereotaxic frame, a glass microelectrode was inserted between L3-L4 spinal cord to a depth of ~400 μ m below the dura, avoiding the posterior spinal arteries. With a stereotaxic injector, 500 nL of viral solution or CTB-555 (1% in PBS; BrainVTA, Wuhan, China) was slowly infused over a period of 5 minutes. The micropipette was left in place for 5 minutes after infusion before being slowly removed.

For knock-down the *Gper1* in CCK⁺ neurons, AAV2/9-CMV-DIO-(EGFP-U6)-shRNA (GPR30)-WPRE-pA (5×10^{12} v.g./mL) or AAV2/9-CMV-DIO-(EGFP-U6)-shRNA (Scramble)-WPRE-pA (5×10^{12} v.g./mL) was injected into the lumbar SDH of *CCK-Cre* mice. For visualization of the CCK⁺ neurons, AAV2/9-CMV-DIO-EGFP-WPRE-pA (5.2×10^{12} v.g./mL) was injected into the lumbar SDH of *CCK-Cre* mice. For anterograde tracing of S1 cortex projections, AAV2/9-hSyn-EGFP-WPRE (1×10^{13} v.g./mL) was injected into the S1 cortex of wild type mice. For visualization of the S1-SDH post-synaptic neurons, AAV2/1-hSyn-CRE-WPRE-pA (1×10^{13} v.g./mL) was injected into the S1 cortex and AAV2/9-EF1 α -DIO-mCherry-WPRE-pA (1×10^{13} v.g./mL) was injected into the lumbar SDH in wild type mice. For visualization of CCK⁺ post-synaptic neurons of S1-SDH projections, AAV2/1-EF1 α -FLP-WPRE-pA (1×10^{13} v.g./mL) was injected into the S1 cortex and AAV2/8-hSyn-Con/Fon-EYFP-WPRE-pA (2×10^{12} v.g./mL) was injected into the lumbar SDH of *CCK-Cre* mice. For chemogenetic manipulation of S1-SDH post-synaptic neurons, AAV2/1-hSyn-CRE-WPRE-pA (1×10^{13} v.g./mL) was injected into the S1 cortex, while AAV2/9-hSyn-DIO-hM3Dq (Gq)-mCherry (3.3×10^{13} v.g./mL; dilution: 1:5) or AAV2/9-hSyn-DIO-hM4Di (Gi)-mCherry (3.3×10^{13} v.g./mL; dilution: 1:5) or AAV2/9-hSyn-DIO-mCherry (3×10^{13} v.g./mL; dilution: 1:5) was injected into the lumbar SDH in wild type mice. All viruses mentioned above were purchased from BrainVTA (Wuhan, China). For visualization of the localization of excitatory interneurons in the SDH, AAV2/9-hSyn-mCaMkIIa-mCherry-WPRE-pA (1×10^{13} v.g./mL; Taitool Bioscience, Shanghai, China) was injected into the lumbar SDH of wild type mice. For structural tests, at least 3 mice were examined and each mouse was examined at least 3 slices. For behavioral tests, at least 6 mice per group were examined. The mice with improper position or expression of virus were excluded.

Chronic constriction injury (CCI)

The CCI-induced neuropathic pain model was employed as previously documented [2]. Mice were lightly anesthetized via inhaled 1.5% Isoflurane. An incision was made on the skin of each mouse, exposing the sciatic nerve. Four ligations with 6-0 chromic silk were loosely tied around the sciatic nerve. Nerve constriction should be minimal until a brief twitch can be observed. In sham mice, the sciatic nerve was exposed without ligation. The animal was allowed to recover from surgery for 2 weeks before behavioral testing.

Behavioral test

Punctate mechanical stimuli (von Frey filaments)

Mice were habituated to opaque cage (7.5×15×15 cm) for 1 hour the day before and 30min immediately prior to testing. Testing was performed using a series of von Frey filaments using the Dixon's Up-down method^[56], beginning with the 0.16 g filament. The 50% paw withdrawal threshold was determined for each mouse on one hind paws. Each filament was gently applied to the plantar surface of the hind paw for 5 seconds or until a response such as a sharp withdraw, shaking or licking of the limb was observed. Between individual measurement, filaments were applied at least 3 minutes after the mice had returned to their initial resting state.

Dynamic mechanical stimuli (Brush)

Each mouse was habituated in an opaque cage (7.5×15×15 cm) for 1 hour the day before and 30 minutes immediately prior to testing. The plantar hind paw was stimulated by light stroking from heel to toe with a paintbrush. A positive response was recorded if the animal lifting, shaking or licking the limb. The application was repeated 10 times with a 3 minutes interval between each stimulation.

Plantar heat test (Hargreaves Method)

Mice were placed in an acrylic chamber on a glass table and allowed to acclimate to the test chamber for 1hour the day before and 30 minutes immediately prior to testing. The thermal paw withdrawal latency was assessed using the plantar test (Ugo Basile Biological Research Apparatus, Gemonio, Italy). While the mouse was in a motionless state, a radiant heat source, which was maintained at 40 W, was applied to the plantar surface of the mouse's paw through the glass plate. The paw withdrawal latency was defined as the time to withdrawal of the hind paw from the heat source, and 15 seconds was used as the cut-off to avoid injury.

Real-time place escape/avoidance test (RT-PEA)

Each mouse was habituated in the test room for 1 hour the day before and 30 minutes immediately prior to testing. The Real-time place escape/avoidance chamber (50×28×32 cm; made with plastic plates that had distinct color with another) was placed on the mesh floor. The tested mouse was placed in a two-chamber box and allowed to explore both chambers without any stimulation (pre-stimulation, 10 minutes); mechanical stimulation by 0.16 g von Frey filament was intermittently delivered whenever the mouse entered or stayed in the preferred chamber, as shown in the pre-stimulation stage (stimulation, 10 minutes); the mouse then freely explored the box without any stimulation (post-stimulation, 10 minutes). The mouse's movements and time stay in preferred chamber were recorded via an ANY-Maze system.

Immunohistochemistry

Mice were deeply anesthetized with 1% pentobarbital sodium solution and then perfused with phosphate-buffered saline (PBS) followed by pre-cooled 4% paraformaldehyde fix solution (PFA). For c-Fos staining, Von Frey filament with the same force (0.16 g) representing light mechanical stimulation was applied to the right hind paw of each group every 30 s for 20 min. Then animals were then perfused with PBS and 4% PFA ninety minutes after Von Frey filament stimulation. Tissue were harvested and post-fixed in PFA at 4°C over night before being dehydrated in 30% sucrose for 2days. Tissues were embedded in Optimal Cutting temperature (OCT) and then cut into 10-30 µm sections placed directly onto slides. Tissue slices were blocked at room temperature for an hour with block solution containing 10% normal donkey serum (NDS), 1% bovine serum albumin BSA and 0.3% triton X-100 in PBS (PBS-T), and then incubated with primary antibodies

diluted in 1% NDS, 1% BSA in PBS-T at 4°C overnight. Sections were washed in PBS and incubated with secondary antibodies at room temperature for 1-2 hours. Slices were washed and covered with Fluoromount-G containing DAPI. All images were taken with an Olympus FV1000 confocal microscope. Antibodies used were as follows: anti-c-Fos (1:1000, guinea pig, Oasis Biofarm, Hangzhou, China), anti-GPR30 (1:500, rabbit, Alomone Labs, Isreal), anti-IBA1 (1:1000, goat, Novusbio, USA), anti-GFAP (1:1000, mouse, Cell Signaling Technology, USA), IB4-FITC (1:1000, Thermofisher, USA), Nissl (1:500, Thermofisher, USA), goat anti-guinea pig IgG-488 (1:500, Oasis Biofarm, Hangzhou, China), donkey anti-rabbit IgG-488 (1:500, Thermofisher, USA), donkey anti-rabbit IgG-555 (1:500, Thermofisher, USA), donkey anti-mouse IgG-488 (1:500, Thermofisher, USA), donkey anti-goat IgG-488 (1:500, Abcam, USA).

In Situ Hybridization

To verify the specificity of transgenic mice, *In situ* hybridization was performed according to the manufacturer's instructions from RNAscope® Multiplex Fluorescent Reagent Kit v2 (Advanced Cell diagnostics, USA) with custom-designed probe for *CCK* (Mm-CCK-C1, Advanced Cell diagnostics, USA). According to the protocols, coronal lumbar spinal sections (10 µm) collected and used for fluorescence in situ hybridization to detect *CCK*⁺ neurons. The slice used to stain RFP primary antibody and *CCK* probe were taken from the -80 °C refrigerator and immediately incubated with pre-cooled 4% PFA for 15 minutes, followed by gradient dehydration (50% ethanol, 70% ethanol, 100% ethanol and 100% ethanol, 5 min for each gradient). The slides were then incubated with RNAscope® hydrogen peroxide at room temperature for 10 minutes, rinsed with distilled water and PBS. Anti-RFP primary antibody (1:1000, rabbit, Rockland, USA) prepared with co-detection diluent (323180, Advanced Cell Diagnostics) was then added to the slices and incubated overnight at 4 °C for subsequent in situ hybridization staining. To characterized the specific expression of GPR30 in SDH, VGAT (Mm-VGAT-C1) and Vglut2 (Mm-Slc17a6-C2) probe were used as described above as well as anti-GPR30 primary antibody (1:250). At least 3 mice were examined and each mouse was examined at least 3 slices in these experiments.

Real-time PCR

Mice lumbar spinal cords or DRG were collected on day 4 after CFA and on day 14 after CCI. Tissues were rapidly collected, frozen in liquid nitrogen and stored at -80°C. RNA was extracted with standard procedures using FastPure Cell/Tissue Total RNA Isolation Kit V2 (Vazyme, Nanjing, China). 500ng of total RNA from each sample was reverse-transcribed with HiScript III RT SuperMix for qPCR (+gDNA wiper) (Vazyme, Nanjing, China). Expression of each mRNA was quantified using ChamQ Universal SYBR qPCR Master Mix (Vazyme, Nanjing, China). The sequences of quantitative PCR primers were as follows: *Gper1*: F: CCTCTGCTACTCCCTCATCG, R: ACTATGTGGCCTGTCAAGGG; GAPDH: F: AAGAAGGTGGTGAAGCAGGCATC, R: CGGCATCGAAGGTGGAAGATG.

Spinal slice preparation and whole-cell recording

Spinal slices were prepared as previously described^[57]. Mice (6-8-week-old, 2 weeks after CCI surgery) were anesthetized with 1% pentobarbital sodium solution and perfused with ice-cold oxygenated (95% O₂ and 5% CO₂) cutting artificial cerebrospinal fluid (ACSF, in mM: 100 sucrose, 63 NaCl, 2.5 KCl, 1.2 NaH₂PO₄, 1.2 MgCl₂, 25 glucose, and 25 NaHCO₃), and the spinal cord was rapidly removed. Transverse spinal cord slices (300 µm, L4 to L6 segment) were prepared using a vibratome (VT1200S, Leica, Germany) and incubated in oxygenated NMDG-ACSF (in mM: 93 NMDG, 2.5 KCl, 1.2 NaH₂PO₄, 30 NaHCO₃, 20 HEPES, 25 Glucose, 5 Na ascorbate, 2 thiourea, 3 Na pyruvate, 10 MgSO₄ and 0.5 CaCl₂, and adjusted to pH 7.4 with HCl) at 34 °C for 15 min. The slices were then transferred to normal ACSF (in mM: 125 NaCl, 2.5 KCl, 1.25 NaH₂PO₄, 1.2 MgCl₂, 2.5 CaCl₂, 25 glucose, and 11 NaHCO₃) at 34 °C for 1h and maintained at room temperature before recording. The slices were transferred to a recording chamber perfused with normal ACSF saturated with 95% O₂ and 5% CO₂.

Whole-cell patch-clamp recordings were performed using a Heka EPC 10 amplifier (Heka Elektronik). Borosilicate glass pipettes with the resistance of 3–5 M Ω were pulled using a horizontal pipette puller (P97, Sutter instruments, USA). The pipettes were filled with cesium-based intracellular fluid (in mM: 100 CsCH₃SO₃, 20 KCl, 10 HEPES, 4 Mg-ATP, 0.3 Tris-GTP, 7 Tris₂-Phosphocreatine, 3 QX-314; pH 7.3, 285–290 mOsm). Targeted whole-cell recordings were made from EGFP expressing neurons in slices taken from *CCK-Cre* mice with virus injection. For spontaneous excitatory post synaptic currents (sEPSCs) recording, the membrane potential was held at –70 mV, and for spontaneous inhibitory post synaptic currents (sIPSCs) recording, the membrane potential was held at +10 mV. Then sEPSCs and sIPSCs were recorded for 100s and analyzed using MiniAnalysis software (Synaptosoft). 5 mice per group were examined in the experiment.

To record AMPA-mediated EPSCs, an electrode placed in the deep laminae of SDH were stimulated at every 15 seconds, and the CCK⁺ neurons were voltage clamped at –70 mV. Meanwhile, the spinal slices were incubated with ACSF containing APV (100 μ M) to block NMDA receptors and bicuculline (20 μ M) and strychnine (0.5 μ M) to block inhibitory synaptic events. AMPAR-mediated EPSCs were recorded for 15 consecutive responses after stable baseline before and after G-1 application (0.1 μ M) to compare to effect of G-1 on AMPAR-mediated EPSCs. 4 mice examined in the experiment.

Statistical analyses

All experiments were randomized. Animals were randomly chosen from multiple cages. For behavior experiments, measurements were taken blinded to condition. All data are reported as mean $\pm$ SEM. The required sample sizes were estimated on the basis of our past experience. Statistical analysis was performed using GraphPad Prism V6. Normal distribution was performed using SPSS V20. For all experiments, $P < 0.05$ was considered to be statistically significant.

Data availability

The data that support the findings of this study are available on request from the corresponding author.

Supplemental Figure Legends

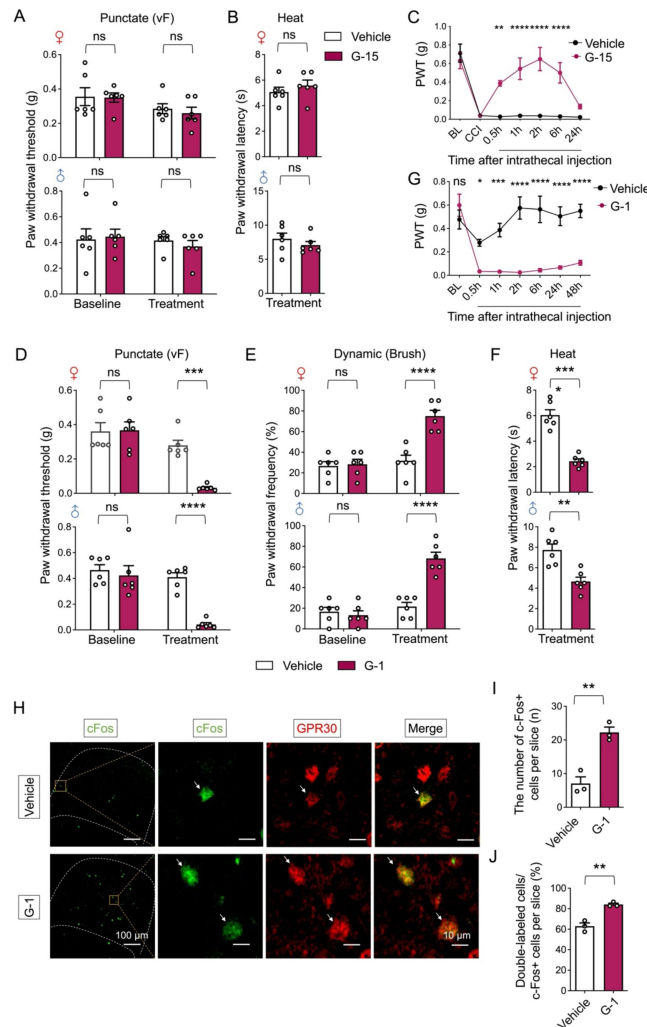


Figure S1.

Spinal inhibition of GPR30 did not change the basic nociception while spinal activation of GPR30 mimicked neuropathic pain symptoms in naïve mice

A, B Behavioral tests of nociception before and after intrathecal injection of G-15 or vehicle in Von Frey tests (**A**) and Heat tests (**B**) in naive mice of both sexes (n = 6 mice for each group). **C** Time course of Von Frey tests after intrathecal injection of G-15 or vehicle in CCI mice (n = 6 mice for each group).

D-F Behavioral tests of nociception before and after intrathecal injection of G-1 or vehicle in Von Frey tests (**D**), Brush tests (**E**) and Heat tests (**F**) in naive mice of both sexes (n = 6 mice for each group).

G Time course of Von Frey tests after intrathecal injection of G-1 or vehicle in naive mice (n = 6 mice for each group).

H Immunohistochemical detection of c-Fos (Green) and GPR30 (Red). Scale bars: 100 μ m. Boxed area of images is enlarged on the right. Scale bars: 10 μ m. White arrows indicate double-positive cells.

I Total number of c-Fos positive neurons in the SDH per section (n = 3 mice for each group, 3-5 pictures were analyzed for each mouse).

J Percentage of c-Fos positive neurons expressing GPR30 (n = 3 mice for each group, 3-5 pictures were analyzed for each mouse).

Data information: in (**A**), ns = not significant (two-way ANOVA with Turkey's multiple comparisons test). In (**B**), ns = not significant (Unpaired Student's t-test). In (**C**), ** $P < 0.01$; **** $P < 0.0001$; ns = not significant (two-way ANOVA with Turkey's multiple comparisons test). In (**D, E**), *** $P < 0.001$; **** $P < 0.0001$; ns = not significant. (two-way ANOVA with Turkey's multiple comparisons test). In (**F**), ** $P < 0.01$; **** $P < 0.0001$ (Unpaired Student's t-test). In (**G**), * $P < 0.05$; *** $P < 0.001$; **** $P < 0.0001$; ns = not significant (two-way ANOVA with Turkey's multiple comparisons test). In (**I, J**), ** $P < 0.01$ (Unpaired Student's t-test). All data are presented as mean $\pm$ SEM.

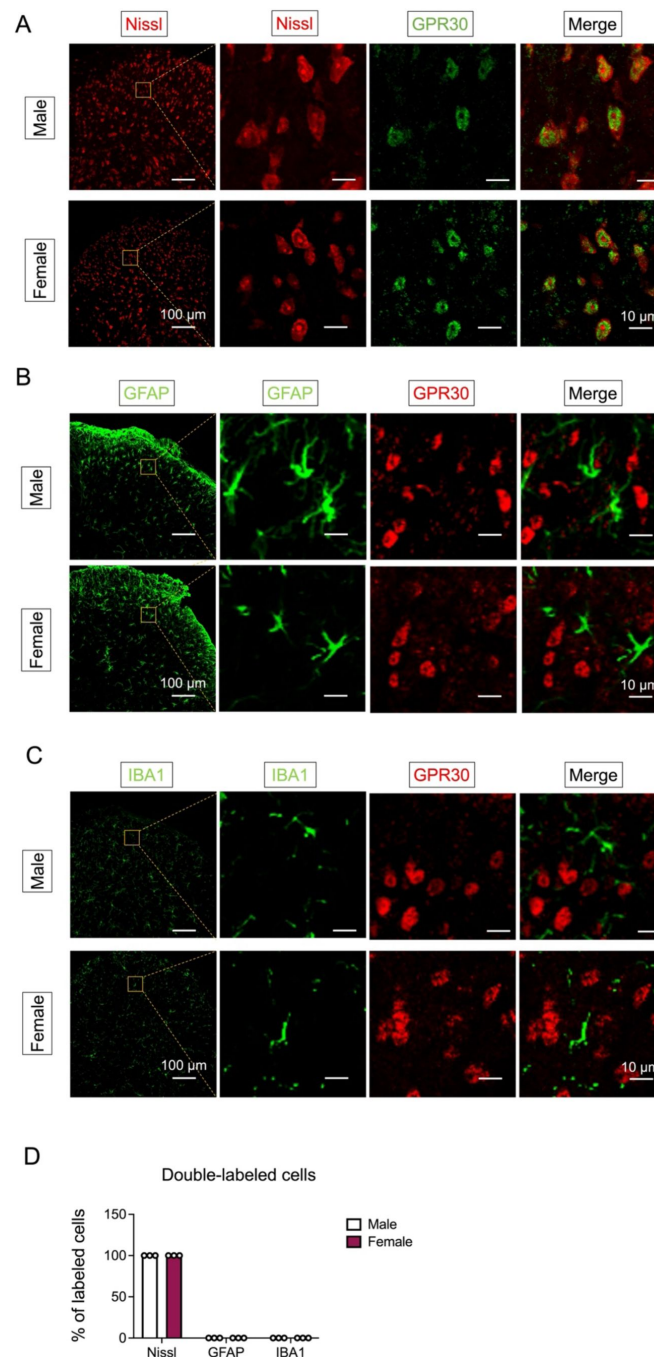


Figure S2.

GPR30 was mainly expressed in the spinal neurons.

A Double-staining of GPR30 (Green) with Nissl (Red) in male (up) and female mice (down). Scale bars: 100 μ m. Boxed area of images is enlarged on the right. Scale bars: 10 μ m.

B Double-staining of GPR30 (Red) with GFAP (Green) in male (up) and female mice (down). Scale bars: 100 μ m. Boxed area of images is enlarged on the right. Scale bars: 10 μ m.

C Double-staining of GPR30 (Red) with IBA1 (Green) in male (up) and female mice (down). Scale bars: 100 μ m. Boxed area of images is enlarged on the right. Scale bars: 10 μ m.

D Percentage of GPR30 expressed in neurons, astrocytes and microglia in both sexes of mice (n = 3 mice for each group, 3 pictures were analyzed for each mouse).

Data information: in (D), data are presented as mean $\pm$ SEM.

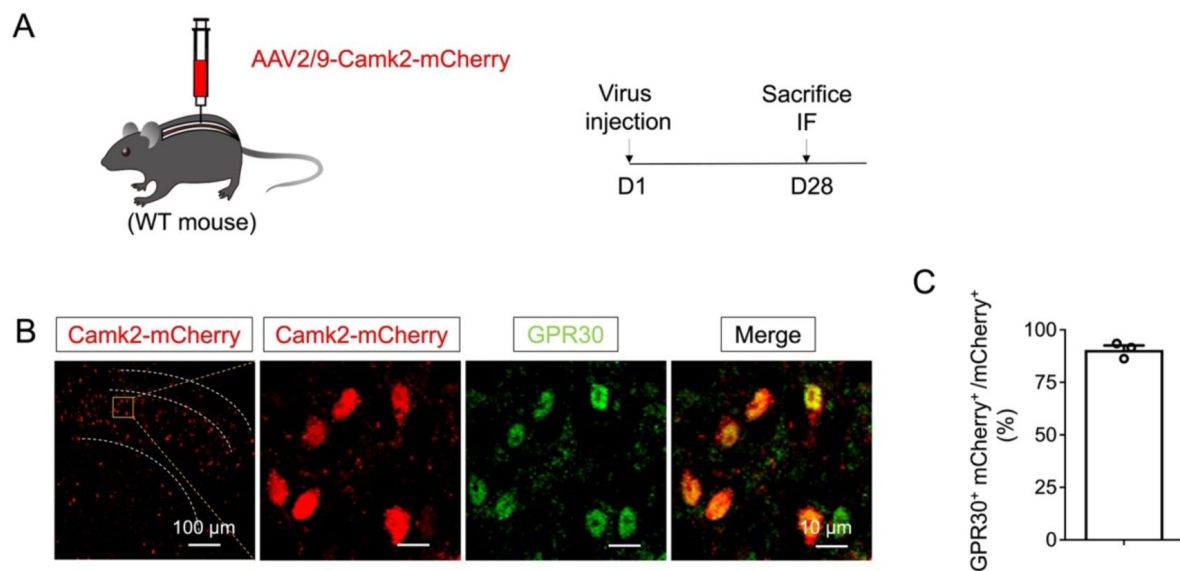


Figure S3.

GPR30 was widely expressed in the Camk2⁺ excitatory interneurons in the SDH.

A Schematic illustration of the strategy to identify the Camk2⁺ excitatory interneurons in spinal dorsal horn by AAV.

B Double staining of Camk2⁺ mCherry⁺ neurons (Red) with GPR30 (Green) by immunohistochemistry. Scale bars: 100 μm. Boxed area of images is enlarged on the right. Scale bars: 10 μm.

C Percentage of Camk2 positive neurons expressing GPR30 in wild mice. (n = 3 mice, 3-4 pictures were analyzed for each mouse)

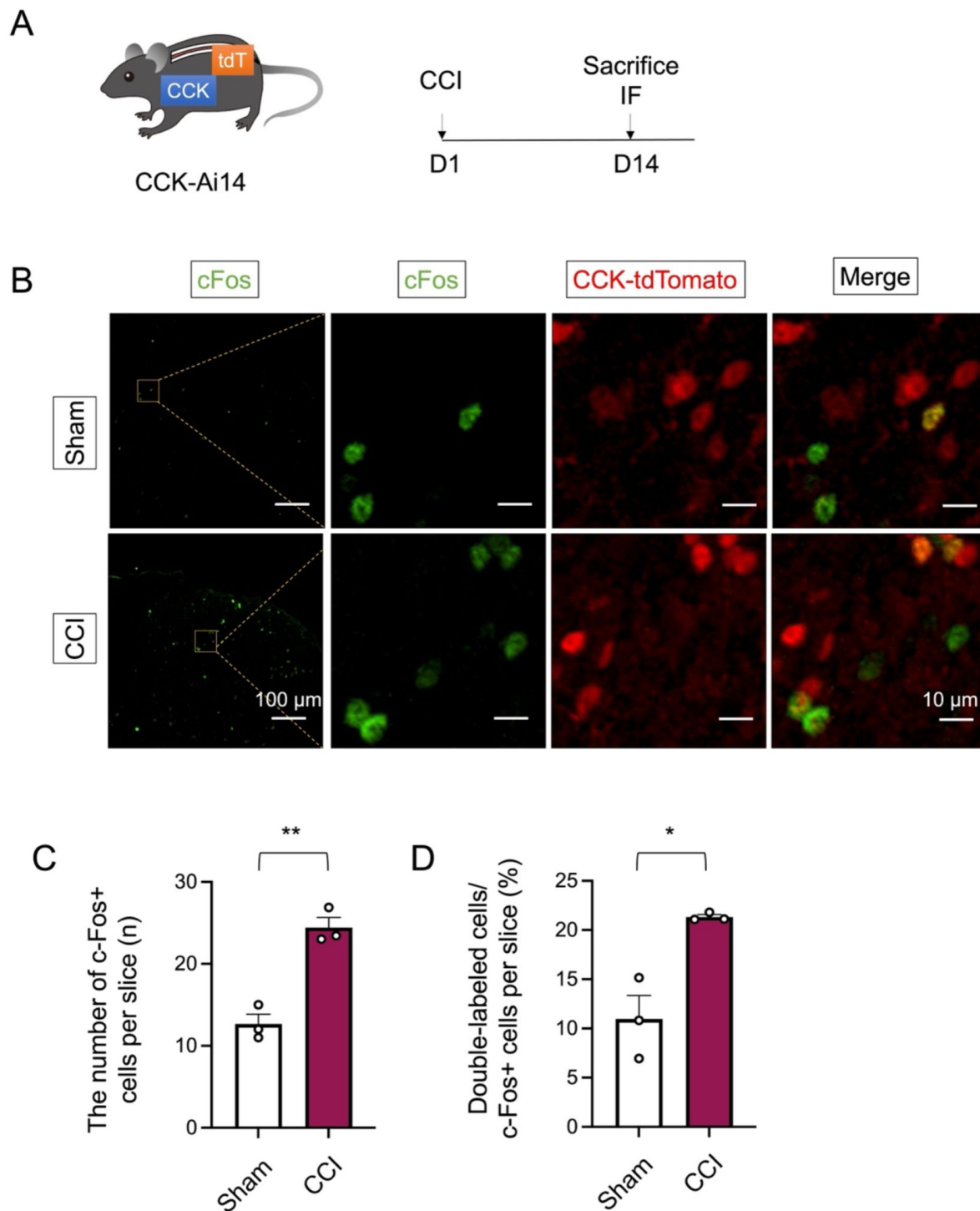


Figure S4.

CCK positive neurons were more activated after CCI.

A Schematic illustration of CCK-Ai14 subjected with CCI.

B Immunofluorescence detection of c-Fos (Green) and CCK-tdTomato (Red). Scale bars: 100 μ m. Boxed area of images is enlarged on the right. Scale bars: 10 μ m.

C Total number of c-Fos positive neurons in the SDH per section (n = 3 mice for each group, 6-8 pictures were analyzed for each mouse).

D Percentage of c-Fos positive neurons expressed on CCK-tdTomato positive neurons (n = 3 mice for each group, 6-10 pictures were analyzed for each mouse).

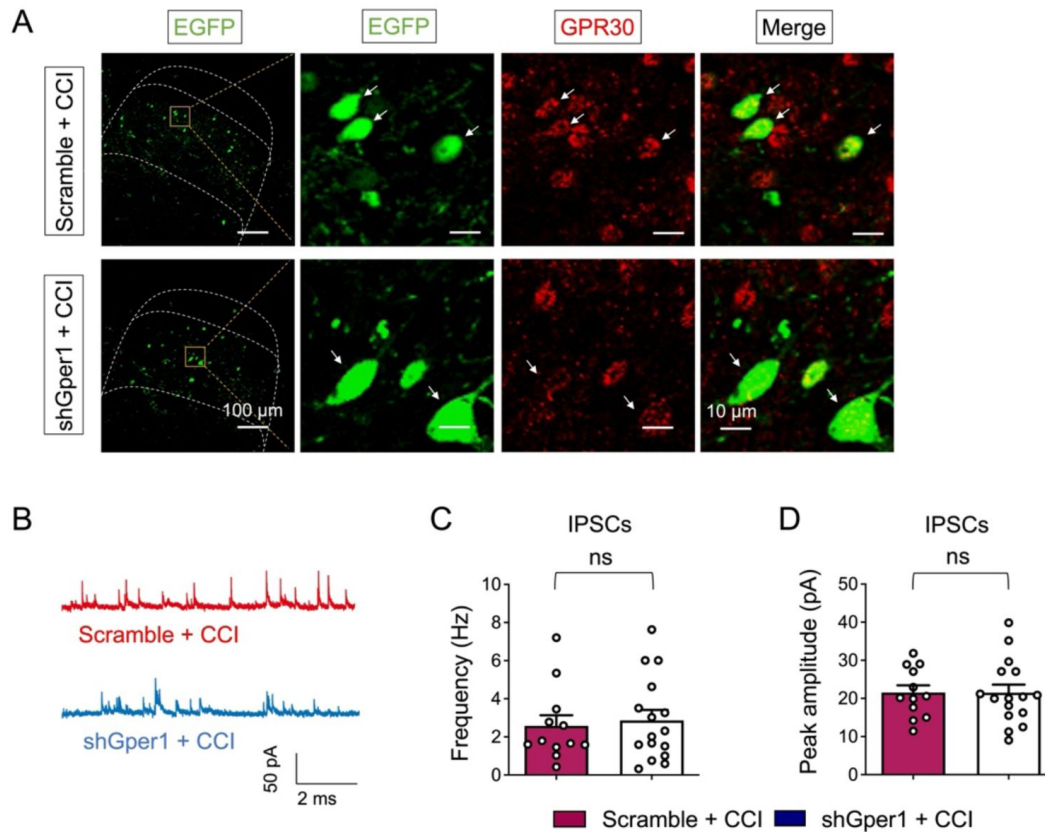


Figure S5.

Knockdown of GPR30 in spinal CCK⁺ neurons did not change the sIPSC in CCI mice.

A Immunohistochemical detection of GPR30 (Red) and shRNA (Green) in mice with virus injection. Note that the intensity of GPR30 fluorescence is less in shGper1-EGFP⁺ cells than that in scramble-EGFP⁺ cells. Scale bars: 100 μ m. Boxed area of image is enlarged on the right. Scale bars: 10 μ m.

B Representative spontaneous IPSCs in CCK-EGFP neurons.

C The frequency of spontaneous IPSCs (n = 12-16 cells from 5 mice).

D The peak amplitude of spontaneous IPSCs (n = 12-16 cells from 5 mice).

Data information: in (**C**, **D**), ns = not significant (Unpaired Student's t-test). All data are presented as mean $\pm$ SEM.

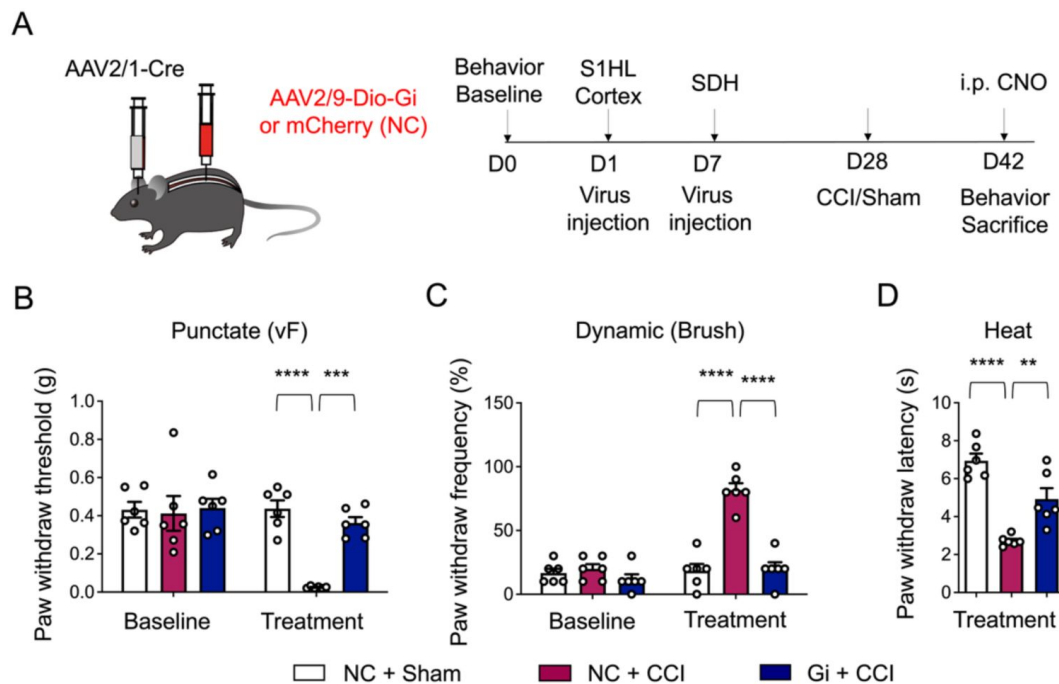


Figure S6.

CCI-induced neuropathic pain was relieved by chemogenetic inhibition of S1-SDH post-synaptic neurons.

A Schematic illustration of the strategy for identifying the post-synaptic neurons of S1HL-SDH projections in SDH (Left) and diagram showing the timeline of AAV2/1 injection in the S1HL cortex, AAV2/9 injection in the spinal dorsal horn, CCI surgery and behavioral tests (Right).

B-D Behavioral tests of basic nociception, 14 days after CCI surgery along with intraperitoneal injection of CNO in Von Frey tests (**B**), Brush tests (**C**) and Heat tests (**D**) in mice ($n = 6$ mice per group).

Data information: In (**B**, **C**), ***P < 0.001; ****P < 0.0001. (two-way ANOVA with Turkey's multiple comparisons test). In (**D**), **P < 0.01; ****P < 0.0001 (one-way ANOVA with Turkey's multiple comparisons test). All data are presented as mean $\pm$ SEM

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Additional information

Conflict of Interest

The authors declare that they have no competing interests.

Author contributions

X.Z.C. and Z.Z.X. conceived the study. Q.C., H.W., S.L.X., J.Q.X., L.H.X., H.L., F.F.Z., conducted the experiments and collected data. Q.C., H.W., and S.L.X. analyzed the data. Y.Y. and H.H.Z. draw the graphic abstract and revise the manuscript. A.G.D., F.X., W.X.Z., L.H.S., Q.X., L.Y.W. and C.C.J. assisted with animal maintenance and provided reagents. Q.C., X.L.Z., Z.Z.X. and X.Z.C. drafted the article. All authors approved the version to be submitted.

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Additional files

Movie 1. Heat test of mice subjected with CCI and spinal viral injection. [🔗](#)

Movie 2. VonFrey test of mice subjected with CCI and spinal viral injection. [🔗](#)

Movie 3. Pre-stimulation phase of RT-PEA test of mice subjected with CCI and shGper1. [🔗](#)

Movie 4. Stimulation phase of RT-PEA test of mice subjected with CCI and shGper1. [🔗](#)

Movie 5. Post-stimulation phase of RT-PEA test of mice subjected with CCI and shGper1. [🔗](#)

Movie 6. Pre-stimulation phase of RT-PEA test of mice subjected with CCI and Scramble. [🔗](#)

Movie 7. *Stimulation phase of RT-PEA test of mice subjected with CCI and Scramble.* [🔗](#)

Movie 8. *Post-stimulation phase of RT-PEA test of mice subjected with CCI and Scramble.* [🔗](#)

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Author information

Qing Chen[#]

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

#These authors contributed equally.

Hui Wu[#]

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

#These authors contributed equally.

Shulan Xie[#]

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China, NANHU Brain-Computer Interface Institute, Hangzhou, China

#These authors contributed equally.

Fangfang Zhu

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

Fang Xu

Department of Anesthesiology, The Second Affiliated Hospital, Chongqing Medical University, Chongqing, China

Qi Xu

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

Lihong Sun

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

Yue Yang

Department of Anesthesiology, Affiliated Hangzhou First People's Hospital, Westlake University School of Medicine, Hangzhou, China

Linghua Xie

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

Jiaqian Xie

Department of Anesthesiology, Sir Run Run Shaw Hospital, School of Medicine, Zhejiang University, Hangzhou, China

Hua Li

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

Ange Dai

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

Wenxin Zhang

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

Luyang Wang

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

Cuicui Jiao

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

Honghai Zhang

Department of Anesthesiology, Affiliated Hangzhou First People's Hospital, Westlake University School of Medicine, Hangzhou, China
ORCID iD: [0000-0003-3530-2060](https://orcid.org/0000-0003-3530-2060)

Xuelong Zhou

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China

Zhen-Zhong Xu

NANHU Brain-Computer Interface Institute, Hangzhou, China, Department of Anesthesiology, First Affiliated Hospital and School of Brain Science and Brain Medicine, Zhejiang University School of Medicine, Hangzhou, China, Liangzhu Laboratory, MOE Frontier Science Center for Brain Science and Brain-machine Integration, State Key Laboratory of Brain-machine Intelligence, NHC and CAMS Key Laboratory of Medical Neurobiology, Zhejiang University, Hangzhou, China

For correspondence: xuzz@zju.edu.cn

Xinzhong Chen

Department of Anesthesia, Women's Hospital, Zhejiang University School of Medicine, Hangzhou, China
ORCID iD: [0000-0001-9219-1681](https://orcid.org/0000-0001-9219-1681)

For correspondence: chenxinz@zju.edu.cn

Editors

Reviewing Editor

Theanne Griffith

University of California, Davis, Davis, United States of America

Senior Editor

John Huguenard

Stanford University School of Medicine, Stanford, United States of America

Reviewer #1 (Public review):

In this manuscript, Chen et al. investigate the role of the membrane estrogen receptor GPR30 in spinal mechanisms of neuropathic pain. Using a wide variety of techniques, they first provide convincing evidence that GPR30 expression is restricted to neurons within the spinal cord, and that GPR30 neurons are well-positioned to receive descending input from the primary sensory cortex (S1). In addition, the authors put their findings in the context the previous knowledge in the field, presenting evidence demonstrating that GPR30 is expressed in the majority of CCK-expressing spinal neurons. Overall, this manuscript furthers our understanding of neural circuitry that underlies neuropathic pain and will be of broad interest to neuroscientists, especially those interested in somatosensation. Nevertheless, the manuscript would be strengthened by additional analyses and clarification of data that is currently presented.

Strengths:

The authors present convincing evidence for expression of GPR30 in the spinal cord that is specific to spinal neurons. Similarly, complementary approaches including pharmacological inhibition and knockdown of GPR30 are used to demonstrate a role for the receptor in driving nerve injury-induced pain in rodent models.

Weaknesses:

Although steps were taken to put their data into the broader context of what is already known about the spinal circuitry of pain, more considerations and analyses would help the authors better achieve their goal. For instance, to determine whether GPR30 is expressed in excitatory or inhibitory neurons, more selective markers for these subtypes should be used over CamK2. Moreover, quantitative analysis of the extent of overlap between GPR30+ and CCK+ spinal neurons is needed to understand the potential heterogeneity of the GPR30 spinal neuron population, and to interpret experiments characterizing descending SI inputs onto GPR30 and CCK spinal neurons. Filling these gaps in knowledge would make their findings more solid.

Revised Manuscript Update:

In their revised manuscript, Chen et al. have added additional data that establishes GPR30 spinal neurons as a population of excitatory neurons, half of which express CCK. These data help to position GPR30 neurons in the existing framework of spinal neuron populations that contribute to neuropathic pain, strengthening the author's findings.

<https://doi.org/10.7554/eLife.102874.2.sa2>

Reviewer #3 (Public review):**Summary:**

The authors convincingly demonstrate that a population of CCK+ spinal neurons in the deep dorsal horn express the G protein coupled estrogen receptor GPR30 to modulate pain sensitivity in the chronic constriction injury (CCI) model of neuropathic pain in mice. Using complementary pharmacological and genetic knockdown experiments they convincingly show that GPR30 inhibition or knockdown reverses mechanical, tactile and thermal hypersensitivity, conditioned place aversion, and c-fos staining in the spinal dorsal horn after CCI. They propose that GPR30 mediates an increase in postsynaptic AMPA receptors after CCI using slice electrophysiology which may underlie the increased behavioral sensitivity. They then use anterograde tracing approaches to show that CCK and GPR30 positive neurons in the

deep dorsal horn may receive direct connections from primary somatosensory cortex. Chemogenetic activation of these dorsal horn neurons proposed to be connected to S1 increased nociceptive sensitivity in a GPR30 dependent manner. Overall, the data are very convincing and the experiments are well conducted and adequately controlled. However, the proposed model of descending corticospinal facilitation of nociceptive sensitivity through GPR30 in a population of CCK+ neurons in the dorsal horn is not fully supported.

Strengths:

The experiments are very well executed and adequately controlled throughout the manuscript. The data are nicely presented and supportive of a role for GPR30 signaling in the spinal dorsal horn influencing nociceptive sensitivity following CCI. The authors also did an excellent job of using complementary approaches to rigorously test their hypothesis.

Weaknesses:

The primary weakness in this manuscript involves overextending the interpretations of the data to still propose a role for corticospinal descending facilitation. While the viral tracing demonstrates a potential connection between S1 and CCK+ or GPR30+ spinal neurons, no direct evidence is provided for S1 in facilitating any activity of these neurons in the dorsal horn.

Comments on the latest version:

The authors did an excellent job addressing many of the critiques raised. Despite acknowledging that a direct functional corticospinal projection to CCK/GPR30+neurons is not supported by the data and revising the title, these claims still persist throughout the manuscript. Manipulating gene expression or the activity of postsynaptic neurons through a trans-synaptic labeling strategy does not directly support any claim that those upstream neurons are directly modulating spinal neurons through the proposed pathway. Indeed they might, but that is not demonstrated here.

<https://doi.org/10.7554/eLife.102874.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

In this manuscript, Chen et al. investigate the role of the membrane estrogen receptor GPR30 in spinal mechanisms of neuropathic pain. Using a wide variety of techniques, they first provide convincing evidence that GPR30 expression is restricted to neurons within the spinal cord, and that GPR30 neurons are well-positioned to receive descending input from the primary sensory cortex (S1). In addition, the authors put their findings in the context of the previous knowledge in the field, presenting evidence demonstrating that GPR30 is expressed in the majority of CCK-expressing spinal neurons. Overall, this manuscript furthers our understanding of neural circuitry that underlies neuropathic pain and will be of broad interest to neuroscientists, especially those interested in somatosensation. Nevertheless, the manuscript would be strengthened by additional analyses and clarification of data that is currently presented.

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pharmacological inhibition and knockdown of GPR30 are used to demonstrate the role of the receptor in driving nerve injury-induced pain in rodent models.

Weaknesses:

Although steps were taken to put their data into the broader context of what is already known about the spinal circuitry of pain, more considerations and analyses would help the authors better achieve their goal. For instance, to determine whether GPR30 is expressed in excitatory or inhibitory neurons, more selective markers for these subtypes should be used over CamK2. Moreover, quantitative analysis of the extent of overlap between GPR30+ and CCK+ spinal neurons is needed to understand the potential heterogeneity of the GPR30 spinal neuron population, and to interpret experiments characterizing descending SI inputs onto GPR30 and CCK spinal neurons. Filling these gaps in knowledge would make their findings more solid.

Thank you very much for your constructive feedback.

In response to your suggestion, we have used more specific markers to distinguish excitatory (VGLUT2) and inhibitory (VGAT) neurons via in situ hybridization. These analyses revealed that GPR30 is predominantly expressed in excitatory neurons of the superficial dorsal horn (SDH), as presented in the Results section (lines 117-120) and in Figure 2A-B.

Additionally, we performed a quantitative analysis to determine the extent of co-localization between GPR30+ and CCK+ neurons. The data were included in the Results (lines 131-132) and Figure 2G.

Reviewer #2 (Public review):

Using a variety of experimental manipulations, the authors show that the membrane estrogen receptor G protein-coupled estrogen receptor (GPER/GPR30) expressed in CCK+ excitatory spinal interneurons plays a major role in the pain symptoms observed in the chronic constriction injury (CCI) model of neuropathic pain. Intrathecal application of selective GPR30 agonist G-1 induced mechanical allodynia and thermal hyperalgesia in male and female mice. Downregulation of GPR30 in CCK+ interneurons prevented the development of mechanical and thermal hypersensitivity during CCI. They also show the up modulation of AMPA receptor expression by GPR30.

Generally, the conclusions are supported by the experimental results. I also would like to see significant improvements in the writing and the description of results.

Methodological details for some of the techniques are rather sparse. For example, when examining the co-localization of various markers, the authors do not indicate the number of animals/sections examined. Similarly, when examining the effect of shGper1, it is unclear how many cells/sections/animals were counted and analyzed.

In other sections, there is no description of the concentration of drugs used (for example, Figure 4H). In Figures 4C-E, there is no indication of the duration of the recordings, the ionic conditions, the effect of glutamate receptor blockers, etc

Some results appear anecdotal in the way they are described. For example, in Figure 5, it is unclear how many times this experiment was repeated.

We sincerely appreciate your valuable feedback and thoughtful recommendations.

To address your concerns regarding methodological transparency, we have added the following details to the revised manuscript:

The number of animals and sections analyzed in co-localization studies.

The number of cells/sections/animals used in each quantification following shGper1 treatment.

The concentrations of drugs administered (e.g., in Figure 4H).

Detailed recording conditions, including duration, ionic composition, and pharmacological conditions (Figures 4C-E).

In addition, we have thoroughly revised the writing throughout the manuscript to enhance clarity and precision in the description of our findings.

Reviewer #3 (Public review):

Summary:

The authors convincingly demonstrate that a population of CCK+ spinal neurons in the deep dorsal horn express the G protein-coupled estrogen receptor GPR30 to modulate pain sensitivity in the chronic constriction injury (CCI) model of neuropathic pain in mice. Using complementary pharmacological and genetic knockdown experiments they convincingly show that GPR30 inhibition or knockdown reverses mechanical, tactile, and thermal hypersensitivity, conditioned place aversion, and c-fos staining in the spinal dorsal horn after CCI. They propose that GPR30 mediates an increase in postsynaptic AMPA receptors after CCI using slice electrophysiology which may underlie the increased behavioral sensitivity. They then use anterograde tracing approaches to show that CCK and GPR30 positive neurons in the deep dorsal horn may receive direct connections from the primary somatosensory cortex. Chemogenetic activation of these dorsal horn neurons proposed to be connected to S1 increased nociceptive sensitivity in a GPR30-dependent manner. Overall, the data are very convincing and the experiments are well conducted and adequately controlled. However, the proposed model of descending corticospinal facilitation of nociceptive sensitivity through GPR30 in a population of CCK+ neurons in the dorsal horn is not fully supported.

Strengths:

The experiments are very well executed and adequately controlled throughout the manuscript. The data are nicely presented and supportive of a role for GPR30 signaling in the spinal dorsal horn influencing nociceptive sensitivity following CCI. The authors also did an excellent job of using complementary approaches to rigorously test their hypothesis.

Weaknesses:

The primary weakness in this manuscript involves overextending the interpretations of the data to propose a direct link between corticospinal projections signaling through GPR30 on this CCK+ population of spinal dorsal horn neurons. For example, even in the cropped images presented, GPR30 is present in many other CCK-negative neurons. Only about a quarter of the cells labeled by the anterograde viral tracing experiment from S1 are CCK+. Since no direct evidence is provided for S1 signaling through GPR30, this conclusion should be revised.

Thank you for your encouraging comments and critical insights.

We fully acknowledge the concern regarding the proposed direct involvement of corticospinal projections in modulating nociceptive behavior via GPR30 in CCK+ neurons.

While our anterograde tracing experiments suggest anatomical overlap, we agree that definitive evidence of functional connectivity is lacking.

Accordingly, we have revised the Abstract, Discussion, and Graphical Abstract to present our findings more cautiously. We now describe our observations as indicating that S1 projections potentially interact with GPR30⁺ spinal neurons, rather than asserting a definitive functional link.

To support this revised interpretation, we performed additional quantitative analyses examining the co-localization among S1 projections, CCK⁺, and GPR30⁺ neurons. Furthermore, we clarified that the chemogenetic activation studies targeted a mixed neuronal population and did not exclusively manipulate CCK⁺ neurons.

These changes aim to better align our conclusions with the presented data and provide a more nuanced framework for future investigations.

Reviewer #1 (Recommendations for the authors):

Major corrections

(1) Figure 2: The authors conclude that GPR30 is mainly expressed in excitatory spinal neurons because they are labeled by a virus with a Camk2 promoter. While there is evidence that Camk2 is specific to excitatory neurons in the brain, based on RNAseq datasets (e.g. Linnarsson Lab, <http://mousebrain.org/adolescent/genesearch.html>) this is less clear cut within the spinal cord. A more direct way to assess the relative expression of GPR30 in excitatory versus inhibitory neurons would be to perform immunohistochemistry or FISH with GPR30/Vglut2/Vgat.

Alternatively, if this observation is not crucial for the overall arch of the story, I recommend the authors eliminate these data, as they do not support the idea that GPR30 is mainly in excitatory neurons.

We thank the reviewer for highlighting this important limitation. To strengthen our conclusion regarding the neuronal identity of GPR30-expressing cells, we performed fluorescent in situ hybridization (FISH) using vGluT2 (marker for excitatory neurons) and VGAT (marker for inhibitory neurons). The results confirmed that GPR30 is predominantly expressed in vGluT2-positive excitatory neurons within the spinal cord. These new data are presented in the revised manuscript (lines 117-120) and shown in Figure 2A-B.

(2) (2a) Figure 2: The authors also report that GPR30 is expressed in most CCK⁺ spinal neurons. A more rigorous way to present the data would be to perform quantification and report the % of CCK neurons that are GPR30.

(2b) More importantly, it is unclear what % of GPR30 neurons are CCK⁺. These types of quantifications would provide useful insights into the heterogeneity of CCK and GPR30 neuron populations, and help align findings of experiments using the behavioral pharmacology using GRP antagonists to the knockdown of Gpr1 in CCK spinal neurons - for instance, does a population of GPR30⁺/CCK⁻ neurons exist? If so, it would be worth discussing what role (if any) that population might play in nerve injury-induced mechanical allodynia.

Understanding the breakdown of GPR30 populations becomes even more relevant when the authors characterize which cell types are targeted by descending projections from S1. It is clear that the vast majority of CCK⁺ neurons that receive descending input from S1 neurons are GPR30⁺, but there are many other GPR30⁺ neurons that do not receive input from SI neurons presented in 5M. Is this simply because only a small fraction of

CCK+/GPR30+ neurons are targeted by descending S1 projections, or could they represent a distinct population of GPR30 neurons?

(2a) We appreciate the suggestion. Quantification showed that approximately 90% of CCK⁺ neurons express GPR30, and about 50% of GPR30⁺ neurons co-express CCK. These data are now provided in the revised Results (lines 131-132) and in Figure 2F-G.

(2b) Indeed, our data reveal that a substantial portion of GPR30⁺ neurons do not co-express CCK. While this study focuses on GPR30 function in CCK⁺ neurons, we recognize the potential relevance of GPR30⁺/CCK⁻ populations. We have addressed this point in the Discussion (lines 303-306):

“However, it should be noted that half of GPR30⁺ neurons are not co-localized with CCK⁺ neurons, and further studies are needed to explore the function of these GPR30⁺/CCK⁻ neurons in neuropathic pain.”

Regarding descending input, our data in Figure 5 show that S1 projections selectively innervate a subset (~30%) of CCK⁺ neurons, most of which co-express GPR30. This suggests that S1-targeted CCK⁺/GPR30⁺ neurons may represent a functionally distinct population. We have added clarification to the revised manuscript, while acknowledging that further studies are needed to elucidate the roles of non-targeted GPR30⁺ neurons.

(3) Throughout the manuscript both male and female mice were used in experiments. Rather than referring to male and female mice as different genders, it would be more appropriate to describe them as different sexes.

As suggested, we have replaced all instances of “gender” with “sex” throughout the revised manuscript.

(4) Figure 5: To increase the ease of interpreting the figure, in panels 5J and 5N, it would be helpful to indicate directly on the figure panel which another marker was assessed in double-labeling analyses.

We have revised Figures 5J and 5N to include clear labels identifying the markers used in double-labeling analyses, to improve interpretability.

Minor corrections:

(1) Line 36, I believe the authors mean to say "GPER/GPR30 in spinal neurons", rather than just "spinal".

Corrected as suggested. The sentence now reads (line 34):

“Here we showed that the membrane estrogen receptor G-protein coupled estrogen receptor (GPER/GPR30) in spinal neurons was significantly upregulated in chronic constriction injury (CCI) mice...”

(2) There are minor grammatical errors throughout the manuscript that interfere with comprehension. Proofreading/editing of the English language use may be beneficial.

We have thoroughly revised the manuscript for clarity and corrected grammatical and syntactic errors to improve readability.

(3) Line 169-170, reads "Known that EPSCs are mediated by glutamatergic receptors like AMPA receptors and several studies have been reported the relationship between GPR30 and AMPA receptor25,29". Rewriting the sentence such that it better describes what the

known relationship is between GPR30 and AMPA would be helpful in setting up the rationale of the experiment in Figure 4.

We have rewritten this section to better clarify the rationale behind the electrophysiological experiments (lines 161-164):

“Given that EPSCs are primarily mediated through glutamatergic receptors such as AMPA receptors, and emerging evidence suggesting that GPR30 enhances excitatory transmission by promoting clustering of glutamatergic receptor subunits, we examined whether GPR30 modulates EPSCs via AMPA receptor-dependent mechanisms.”

(4) Line 198-199 "Then we explored the possible connections among GPR30, S1-SDH projections and CCK+ neuron." In the context of spinal circuitry, "connections" may raise the expectation that synaptic connectivity will be evaluated. What I think best describes what the authors investigated in Figure 5 is the "relationship" between GPR30, S1-SDH projections, and CCK+ neurons.

We have revised the sentence accordingly (lines 184-186):

“Building on previous findings suggesting a functional interaction between S1-SDH projections and spinal CCK⁺ neurons, our current study aimed to further elucidate the structural relationship among GPR30, S1-SDH projections, and CCK⁺ neurons.”

(5) Figure 5: To increase the ease of interpreting the figure, in panels 5J and FN, it would be helpful to indicate directly on the figure panel which other marker was assessed in double-labeling analyses.

We have added direct labels to figure panels to clarify double-labeled analyses in the revised Figure 5J and 5N.

Reviewer #2 (Recommendations for the authors):

(1) Can the authors provide more detail about the distribution of CCK+ cells in the spinal cord and, in particular, the localization of double-stained (CCK/cfos) neurons?

We thank the reviewer for this suggestion. To better characterize the distribution of CCK⁺ neurons within the spinal dorsal horn (SDH), we performed immunostaining in CCK-tdTomato mice using lamina-specific markers: CGRP (lamina I), IB4 (lamina II), and NF200 (lamina III–V). Our results demonstrate that CCK⁺ neurons are primarily localized in the deeper laminae of the SDH. These findings are now described in the revised Results (lines 126–129) and shown in Figure 2E.

In addition, we conducted c-Fos immunostaining in CCK-Ai14 mice and found increased activation of CCK⁺ neurons following CCI. This supports the involvement of CCK⁺ neurons in neuropathic pain. These data are included in the Results (lines 129–131) and Supplementary Figure S4.

(2) Figure 2A. There is no formal quantification of the percentage of TdTomato+ neurons that are also CCK+. The description of these results is insufficient.

We appreciate this point and have revised the description of Figure 2A accordingly. To strengthen our analysis, we conducted additional FISH experiments with vGluT2 and VGAT probes. Quantification revealed that GPR30 is predominantly expressed in excitatory neurons (approximately 60%). These data are shown in the revised Results (lines 117-119) and Figures

2A-B and S3. This supports our conclusion that GPR30 is largely localized to excitatory spinal interneurons.

(3) *Figure 4H. What is the evidence that these are AMPA-mediated currents? This is not explained in the text.*

Thank you for raising this point. We now provide detailed experimental procedures to clarify that the recorded EPSCs are AMPA receptor-mediated. Specifically, spinal slices from CCK-Cre mice were used, and excitatory postsynaptic currents were recorded in the presence of APV (100 μ M, NMDA receptor blocker), bicuculline (20 μ M, GABA_A receptor blocker), and strychnine (0.5 μ M, glycine receptor blocker), ensuring that the observed currents were AMPA-dependent. These methodological details are now clearly described in the revised Results (lines 165–173) and supported by prior literature (Zhang et al., J Biol Chem 2012; Hughes et al., J Neurosci 2010).

(1) Yan Zhang, Xiao Xiao, Xiao-Meng Zhang, Zhi-Qi Zhao, Yu-Qiu Zhang (2012). Estrogen facilitates spinal cord synaptic transmission via membrane-bound estrogen receptors: implications for pain hypersensitivity. J Biol Chem. Sep 28;287(40):33268-81.

(2) Ethan G Hughes, Xiaoyu Peng, Amy J Gleichman, Meizan Lai, Lei Zhou, Ryan Tsou, Thomas D Parsons, David R Lynch, Josep Dalmau, Rita J Balice-Gordon (2010). Cellular and synaptic mechanisms of anti-NMDA receptor encephalitis. J Neurosci. 2010 Apr 28;30(17):5866-75.

(4) *What is the signaling mechanism leading to a larger amplitude of currents after G-1 infusion?*

We thank the reviewer for this important question. G-1 is a selective agonist for GPR30. Based on previous studies by Luo et al. (2016), we speculate that activation of GPR30 may increase the clustering of glutamatergic receptor subunits at postsynaptic sites, thereby enhancing AMPA receptor-mediated currents. While our current study did not directly address the intracellular signaling cascade, we have incorporated this mechanistic speculation in the Discussion.

Jie Luo, X.H., Yali Li, Yang Li, Xueqin Xu, Yan Gao, Ruoshi Shi, Wanjun Yao, Juying Liu, Changbin Ke (2016). GPR30 disrupts the balance of GABAergic and glutamatergic transmission in the spinal cord driving to the development of bone cancer pain. Oncotarget 7, 73462-73472. 10.18632/oncotarget.11867.

(5) *Figure 4I. Please include error bars.*

We have revised Figure 4I to include error bars, as requested.

(6) *Line 198. What is the evidence that AAV2/1 EF1 α FLP is an anterograde trans monosynaptic marker?*

We thank you for this request. AAV2/1 has been widely used for anterograde monosynaptic tracing based on its properties (Wang et al., Nat Neurosci 2024; Wu et al., Neurosci Bull 2021): (1) it infects neurons at the injection site and undergoes active anterograde transport; (2) newly assembled viral particles are released at synapses and infect postsynaptic partners; (3) in the absence of helper viruses, the spread halts at the first synapse, ensuring monosynaptic restriction. We have elaborated on this in the revised manuscript (line 198), citing Wang et al. (Nat Neurosci 2024) and Wu et al. (Neurosci Bull 2021).

(1) Hao Wang, Qin Wang, Liuzhe Cui, Xiaoyang Feng, Ping Dong, Liheng Tan, Lin Lin, Hong Lian, Shuxia Cao, Huiqian Huang, Peng Cao, Xiao-Ming Li (2024). A molecularly defined

amygdalaindependent tetra-synaptic forebrain-tohindbrain pathway for odor-driven innate fear and anxiety. *Nat Neurosci.* 2024 Mar;27(3):514-526.

(2) Zi-Han Wu, Han-Yu Shao, Yuan-Yuan Fu, Xiao-Bo Wu, De-Li Cao, Sheng-Xiang Yan, Wei-Lin Sha, Yong-Jing Gao, Zhi-Jun Zhang (2021). Descending Modulation of Spinal Itch Transmission by Primary Somatosensory Cortex. *Neurosci Bull.* 2021 Sep;37(9):1345-1350.

(7) Figure 5G. I do not understand the logic of this experiment. A Cre AAV is injected in the S1 cortex. Why should this lead to the expression of tdTomato on a downstream (postsynaptic?) neuron? The authors should quote the literature that supports this anterograde transsynaptic transport.

We appreciate this question. As described in previous studies (e.g., Wu et al., *Neurosci Bull* 2021), AAV2/1-Cre injected into the S1 cortex leads to Cre expression in projection targets due to transsynaptic anterograde transport. Subsequent injection of a Cre-dependent AAV (AAV2/9-DIO-mCherry) into the spinal cord enables specific labeling of postsynaptic neurons that receive input from S1. We have clarified this mechanism in line 206 and provided the appropriate citation.

Zi-Han Wu, Han-Yu Shao, Yuan-Yuan Fu, Xiao-Bo Wu, De-Li Cao, Sheng-Xiang Yan, Wei-Lin Sha, Yong-Jing Gao, Zhi-Jun Zhang (2021). Descending Modulation of Spinal Itch Transmission by Primary Somatosensory Cortex. *Neurosci Bull.* 2021 Sep;37(9):1345-1350.

(8) The same question arises when interpreting the results obtained in Figure 6.

We thank the reviewer for the question, and we have addressed it in point (7).

(9) Line 257. How do the authors envision that estrogen would change its modulation of GPR30 under basal and neuropathic conditions? Is there any evidence for this speculation?

We thank the reviewer for raising this thoughtful question. In the current study, we focused on pharmacologically manipulating GPR30 activity via its selective agonist and antagonist. We did not directly investigate how endogenous estrogen regulates GPR30 under physiological and neuropathic states. We have recognized this limitation and highlighted the need for future research to investigate this regulatory mechanism.

(10-20) In my opinion, the entire manuscript needs a careful revision of the English language. While one can follow the text, it contains numerous grammatical and syntactic errors that make the reading far from enjoyable. I am highlighting just a few of the many errors.

We appreciate the reviewer's honest assessment. The manuscript has undergone thorough language editing by a native English speaker to correct grammatical errors, improve clarity, and enhance overall readability. We also restructured several sections, particularly the Discussion, to improve logical flow.

(21) The discussion of results is a bit disorganized, with disconnected sentences and statements, and somewhat repetitive. For example, lines 303 to 306 lack adequate flow. It is also quite long and includes general statements that add little to the discussion of the new findings (lines 326-333).

We agree and have revised the Discussion extensively. Disconnected or repetitive sentences (e.g., lines 303-306, 326-333) have been removed or rewritten. For instance, we added a new transitional paragraph (lines 307-311) to improve flow:

“Abnormal activation of neurons in the SDH is a key contributor to hyperalgesia, and enhanced excitatory synaptic transmission is a major mechanism driving increased neuronal excitability. Therefore, we evaluated excitatory postsynaptic currents (EPSCs) and observed increased amplitudes in CCK⁺ neurons following CCI, suggesting elevated excitability in these neurons.”

We also removed redundant generalizations to maintain a focused discussion of our novel findings.

Reviewer #3 (Recommendations for the authors):

(1) What is the distribution of GPR30 throughout the spinal cord and DRG? The authors demonstrate that this can overlap with a CCK⁺ population, but there are many GPR30⁺ and CCK negative neurons, even in the cropped images presented. It would be helpful to quantify the colocalization with CCK.

We thank the reviewer for this important point. As shown in the revised manuscript, GPR30 is expressed in both the spinal cord and dorsal root ganglia (DRG). However, our updated data (Figure 1B) demonstrate that Gper1 mRNA levels in the DRG are not significantly altered after CCI, suggesting a limited involvement of DRG GPR30 in neuropathic pain. These results are described in the revised Results (line 94).

Regarding spinal co-expression, we performed a detailed quantification. Approximately 90% of CCK⁺ neurons express GPR30, while about 50% of GPR30⁺ neurons are CCK⁺. These co-localization results are now included in the revised Results and presented in Figure 2G.

(2) It is clear that CCI and GPR30 influence excitatory synaptic transmission in CCK⁺ neurons. However, these experiments do not fully support the authors' claims of a postsynaptic upregulation of AMPARs. Comparing amplitudes and frequencies of spontaneous EPSCs cannot necessarily distinguish a pre- vs postsynaptic change since some of these EPSCs can arise from spontaneous action potential firing. I suggest revising this conclusion.

We appreciate these insightful comments. We fully agree that our data from spontaneous EPSC recordings (sEPSCs) in CCK⁺ neurons are not sufficient to distinguish between pre- and postsynaptic mechanisms, as sEPSCs may include spontaneous presynaptic activity. Therefore, we have revised the text throughout the manuscript to avoid overstating conclusions related to postsynaptic AMPA receptor upregulation.

(3) What is the rationale for the evoked EPSC experiments from electrical stimulation in "the deep laminae of SDH?" I do not think that this experiment can rule out a presynaptic contribution of GPR30 to the evoked responses, particularly if these are Gs-coupled at presynaptic terminals. Paired-pulse stimulations could help answer this question, otherwise, alternative interpretations, also related to the point above, should be provided.

We thank the reviewer for this thoughtful critique. Indeed, electrical stimulation of the deep SDH laminae does not exclude presynaptic involvement, especially considering that GPR30 is a G protein-coupled receptor (GPCR) and could act presynaptically. We agree that paired-pulse ratio (PPR) analysis would be more informative in distinguishing pre- from postsynaptic effects, but this was not performed due to technical limitations in our current experimental setup.

Accordingly, we have revised our interpretations in both the Results and Discussion to acknowledge that our data do not rule out presynaptic contributions. We now state that

GPR30 activation enhances EPSCs in CCK⁺ neurons, while further studies are needed to dissect the precise site of action.

(4) I appreciate the challenging nature of the trans-synaptic viral labeling approaches, but the chemogenetic and Gpr knockdown experiments do not selectively target this CCK⁺ population of deep dorsal horn neurons. The data are clear that each of these components (descending corticospinal projections, CCK neurons, and GPR30) can modulate nociceptive hypersensitivity, but I do not agree with the overall conclusion that each of are directly linked as the authors propose. I recommend revising the overall conclusion and title to reflect the convincing data presented.

We thank the reviewer for this critical observation. We agree that while our data show functional roles for descending cortical input, CCK⁺ neurons, and GPR30 in modulating pain hypersensitivity, the evidence does not establish a definitive direct circuit integrating all three components.

In response, we have revised our conclusions to reflect this limitation. Specifically, we avoided claiming a direct functional link among S1 projections, CCK⁺ neurons, and GPR30. Instead, we now propose that GPR30 modulates neuropathic pain primarily through its action in CCK⁺ spinal neurons, with potential involvement of descending facilitation from the somatosensory cortex.

Additionally, we have revised the manuscript title to better reflect our mechanistic focus: “GPR30 in spinal CCK-positive neurons modulates neuropathic pain.”

Minor Corrections

(1) The authors should refer to mice by sex, not gender.

Corrected throughout the manuscript.

(2) Page 9, line 195: "significantly" is used to refer to co-localization of 28.1%. What is this significant to?

We have revised the sentence to accurately describe the observed percentage, without implying statistical significance:

“Our co-staining results revealed that a high proportion of CCK⁺ S1-SDH postsynaptic neurons expressed GPR30” (line 198-199).

(3) I recommend modifying some of the transition phrases like "by the way," "what's more," and "besides".

All informal expressions have been replaced with academic alternatives including “Furthermore,” “Additionally,” and “Moreover.”

(4) Additional guides to mark specific laminae in the dorsal horn would be useful.

We added immunostaining with laminar markers (CGRP for lamina I and NF200 for lamina III–V), and these data are now shown in Figure 2E and described in the Results (lines 126-129).

(5) Page 5, line 115: immunochemistry should be immunohistochemistry.

Corrected as suggested.

(6) Page 6, line 136: *"Confirming the structural connnections" was not demonstrated here. Perhaps co-localization between GPR30 and CCK+.*

The text was revised to "To functionally interrogate GPR30 and CCK⁺ neurons in neuropathic pain..." (line 133).

(7) Page 8, line 166: *unsure what "took and important role" means.*

This phrasing was corrected for clarity and replaced with an accurate scientific description.

(8) Page 8, line 168: *"IPSCs of spinal CCK+ neurons" implies that they are sending inhibitory inputs.*

We revised the term to "EPSCs" to correctly reflect excitatory synaptic currents in CCK⁺ neurons.

(9) Page 8, line 169: *"Known that EPSCs" is missing an introductory phrase.*

The sentence was rewritten to include an appropriate introductory clause (lines 161–164):

"Given that EPSCs are primarily mediated through glutamatergic receptors such as AMPA receptors..."

(10) Page 10, line 227 and 228: *"adequately" and "sufficiently" should be adequate and sufficient.*

We corrected these terms to the proper adjective forms: "adequate" and "sufficient" (lines 224–225).

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